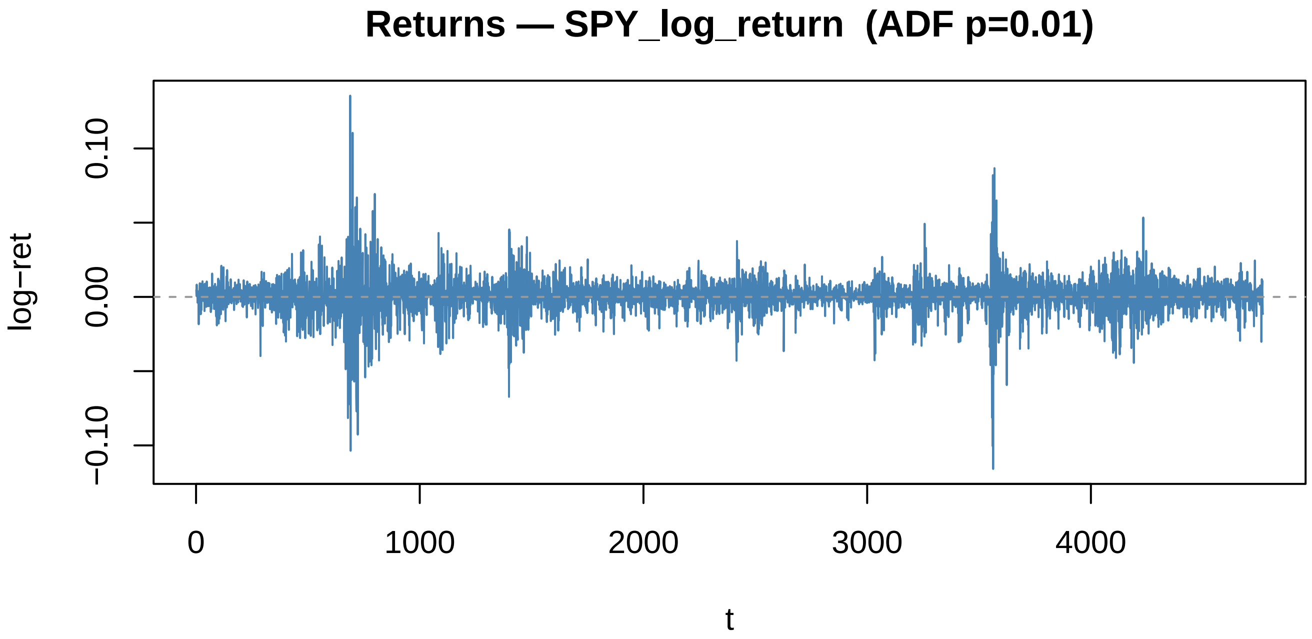
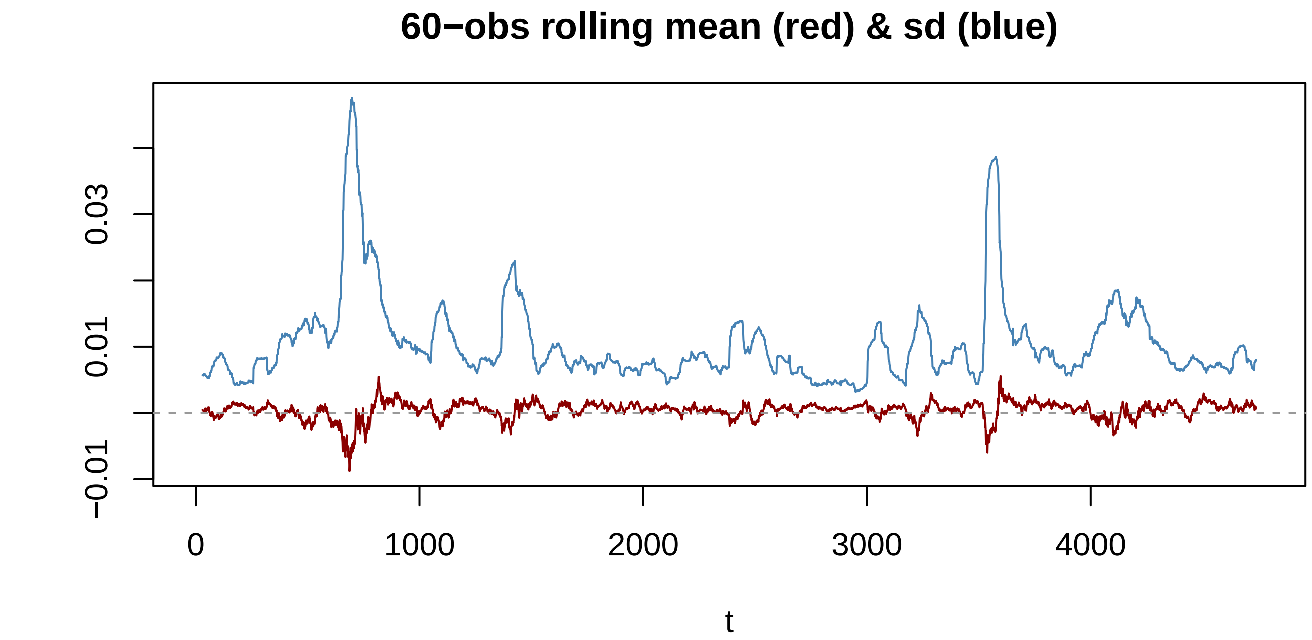
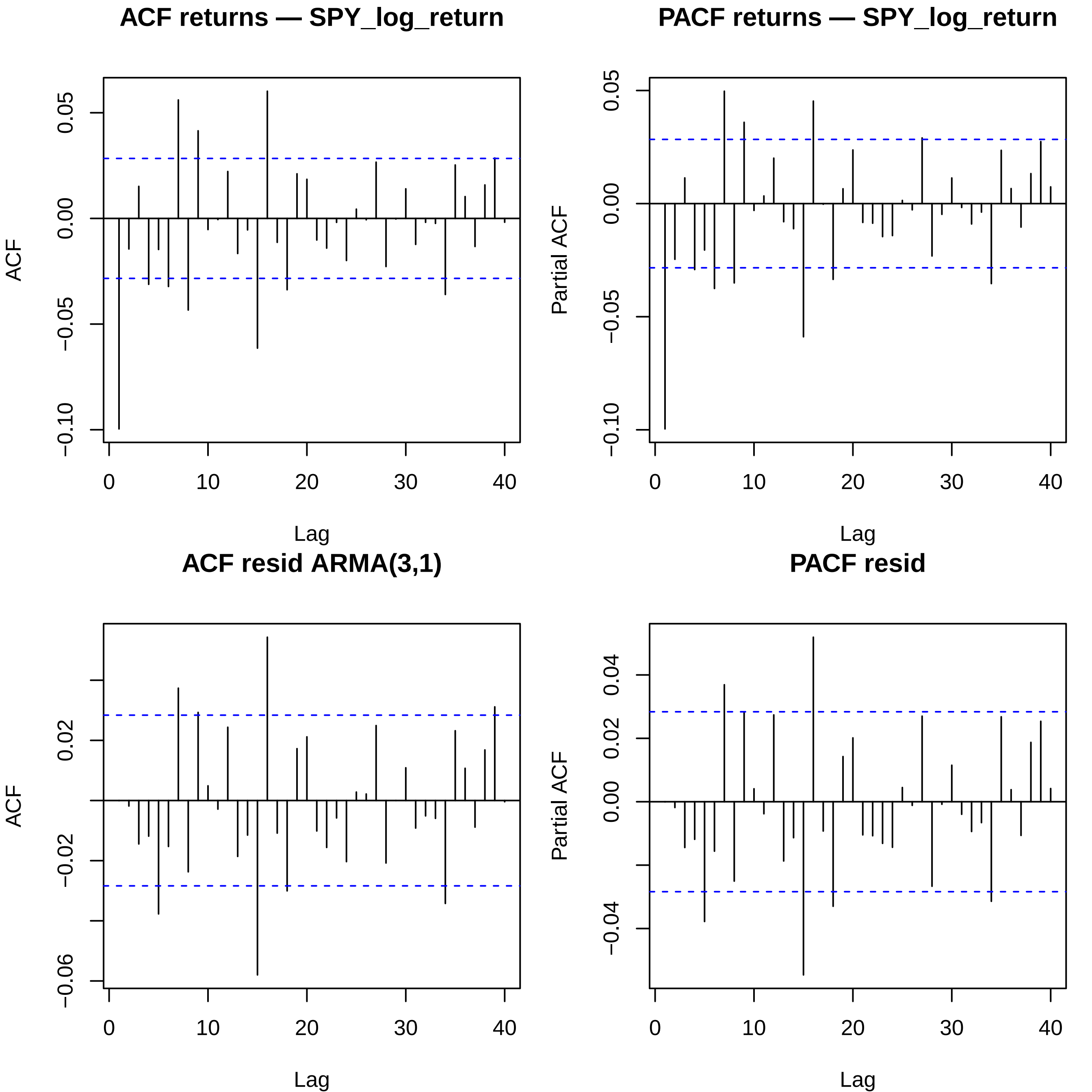
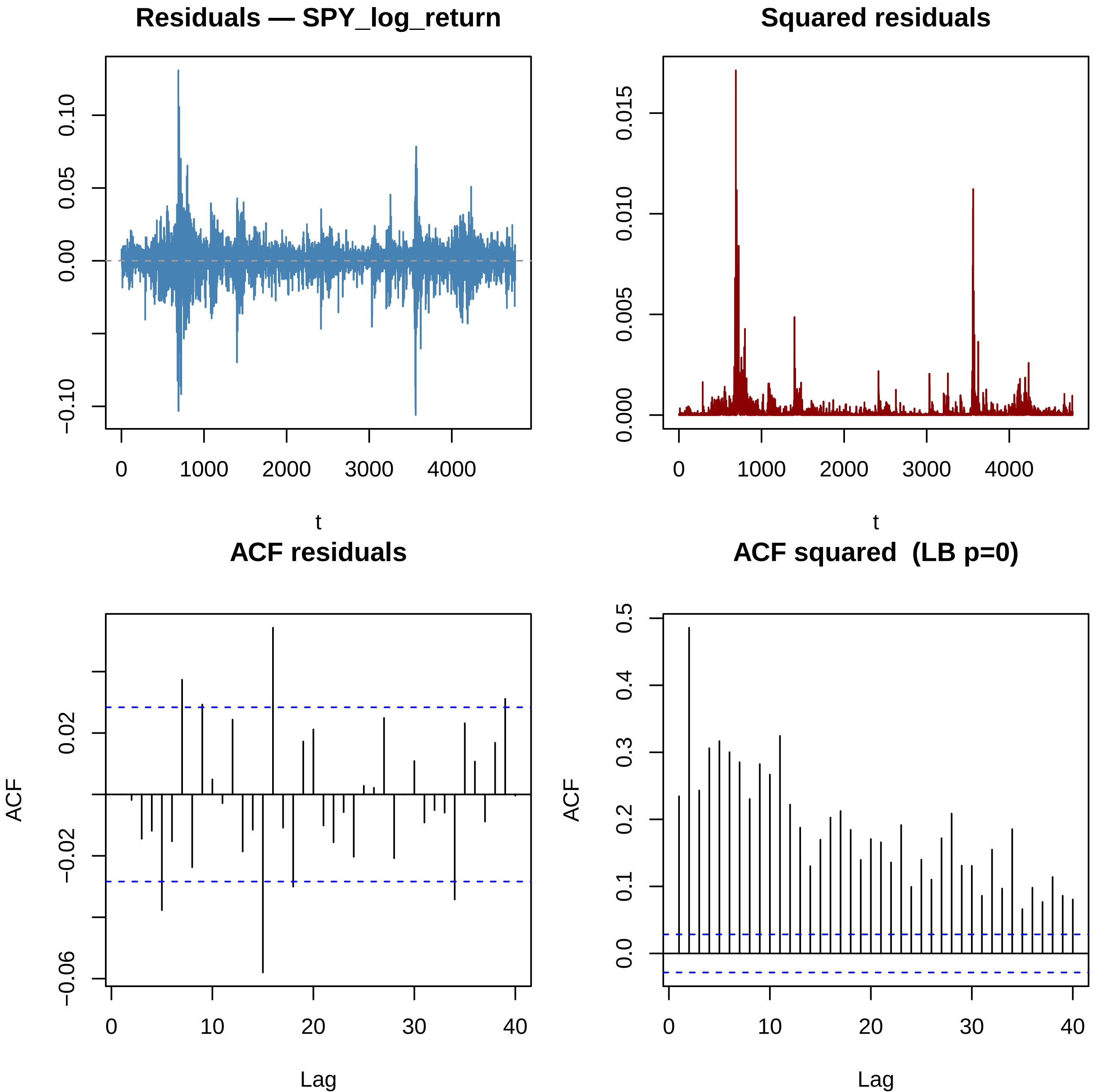
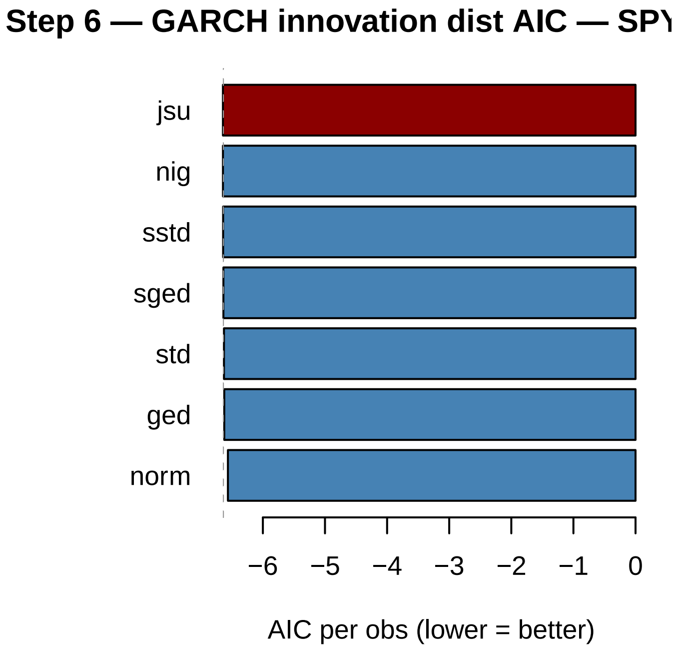

Returns — SPY_log_return (ADF p=0.01)
log−ret
0.10
0.00
−0.10
0
1000
2000
3000
4000
t

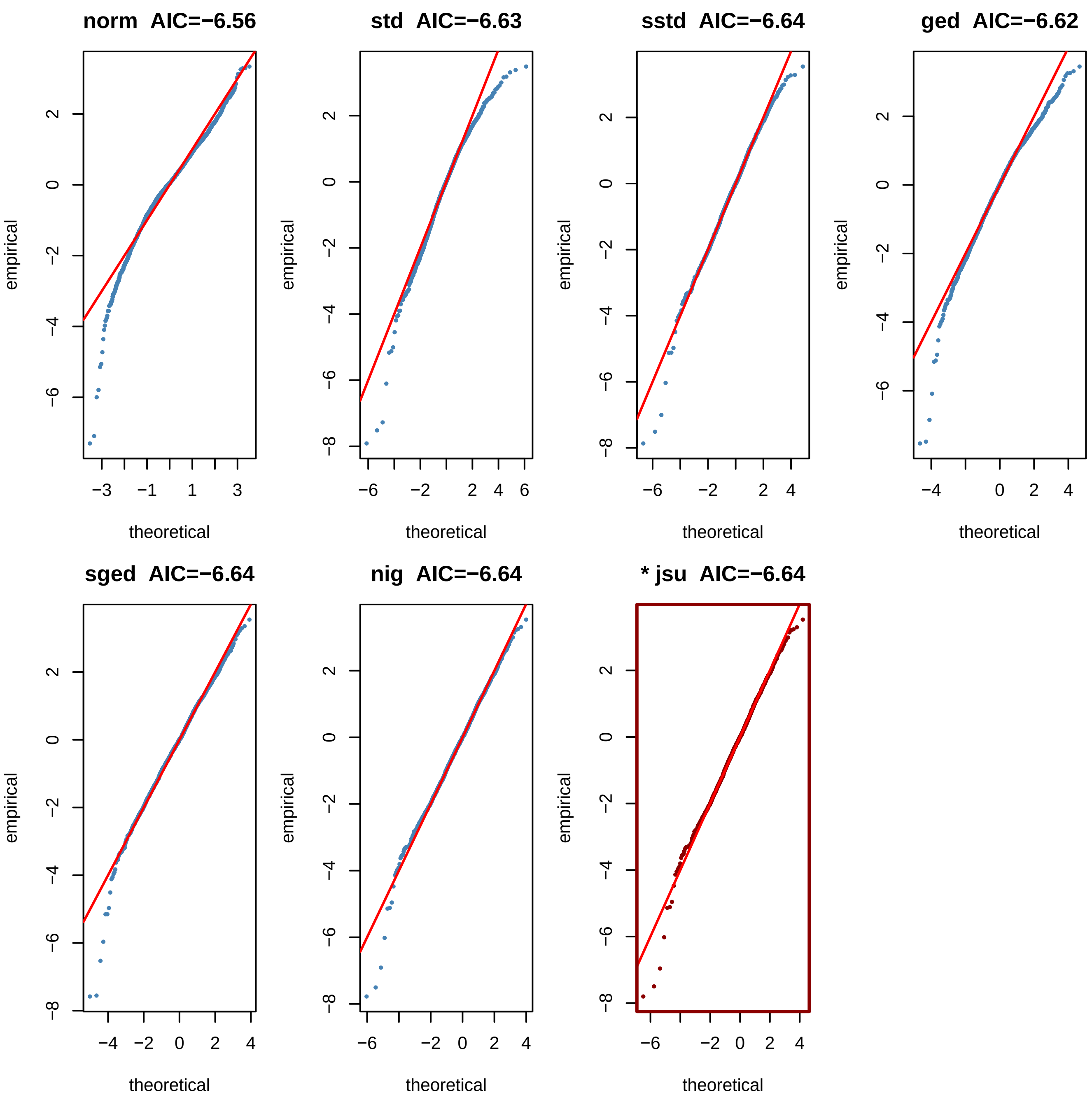

60−obs rolling mean (red) & sd (blue)
0.03
0.01
−0.01
0
1000
2000
3000
4000
t



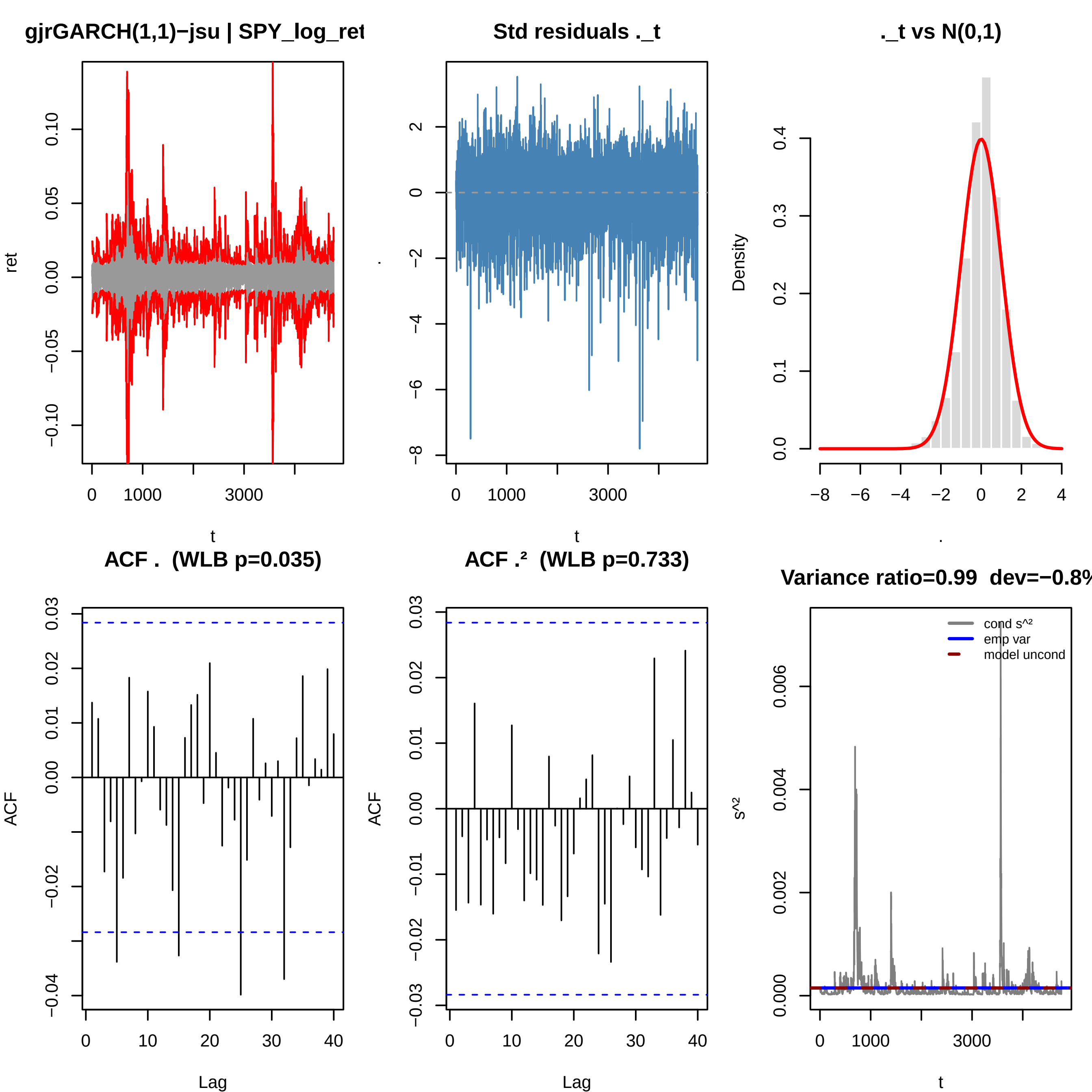
ACF returns — SPY_log_return
PACF returns — SPY_log_return
ACF resid ARMA(3,1)
PACF resid
ACF
Partial ACF
Lag
0
10
20
30
40
0.05
0.00
−0.05
−0.10
0.02
−0.02
−0.06
0.04
−0.04

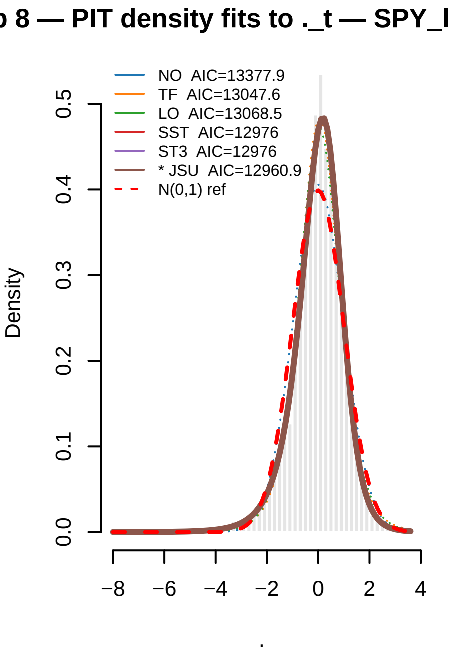

Residuals — SPY_log_return
0.10
0.05
0.00
−0.10
0
1000
2000
3000
4000
t
Squared residuals
0.015
0.010
0.005
0.000
0
1000
2000
3000
4000
t
ACF residuals
0.02
−0.02
−0.06
ACF
0
10
20
30
40
Lag
ACF squared (LB p=0)
0.5
0.4
0.3
0.2
0.1
0.0
ACF
0
10
20
30
40
Lag

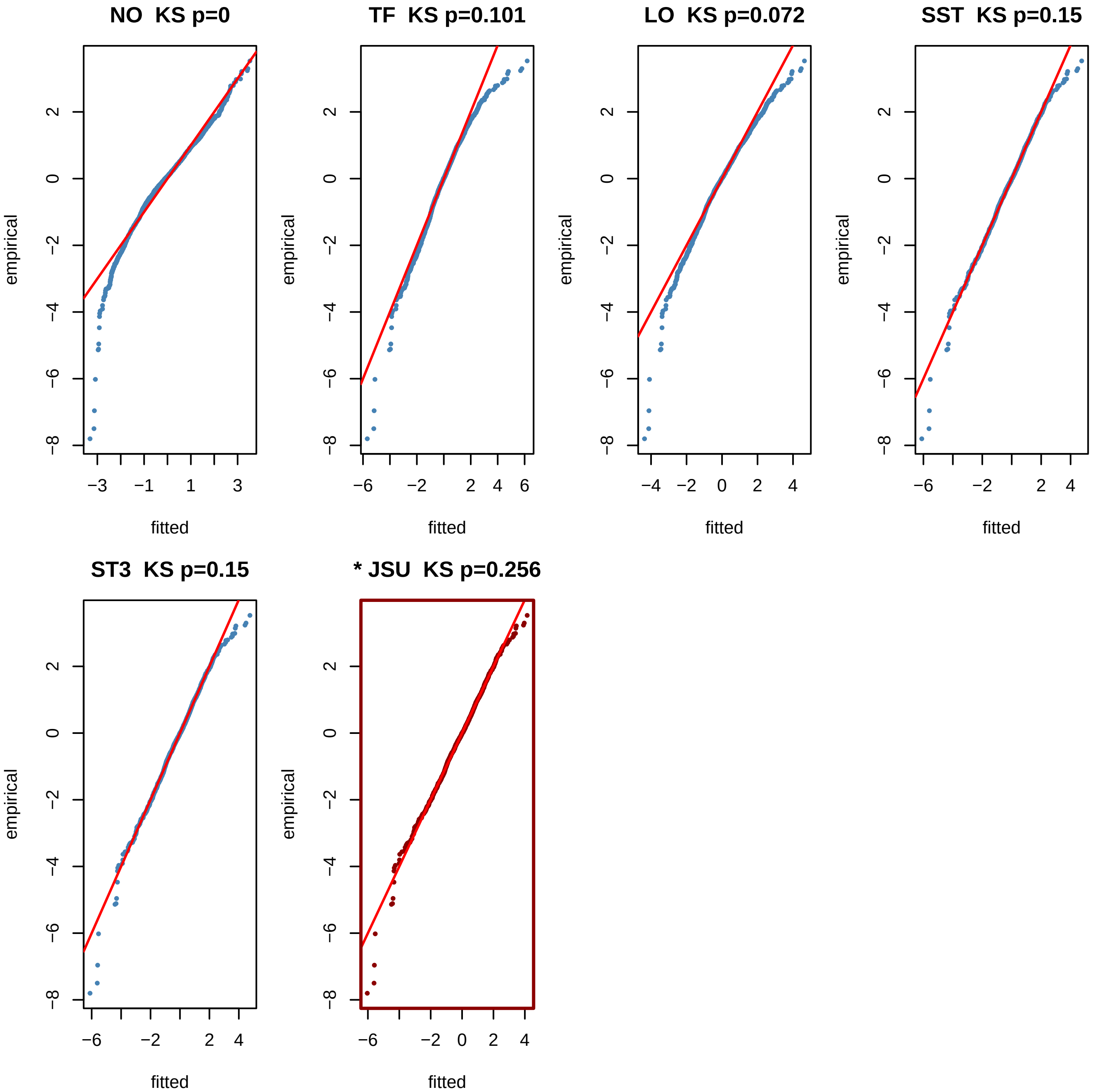

Step 6 — GARCH innovation dist AIC — SPY
jsu
nig
sstd
sged
std
ged
norm
−6
−5
−4
−3
−2
−1
0
AIC per obs (lower = better)

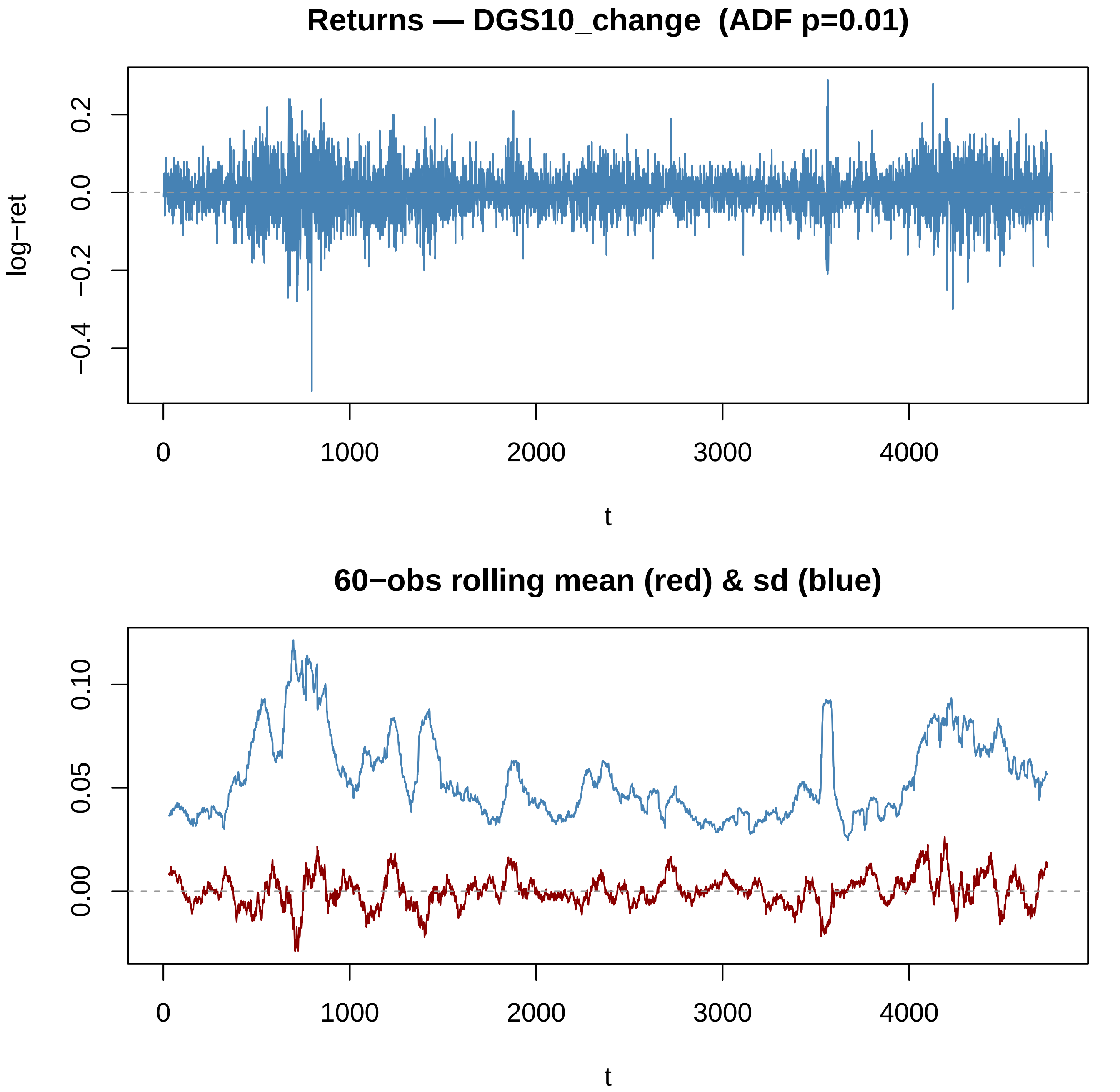


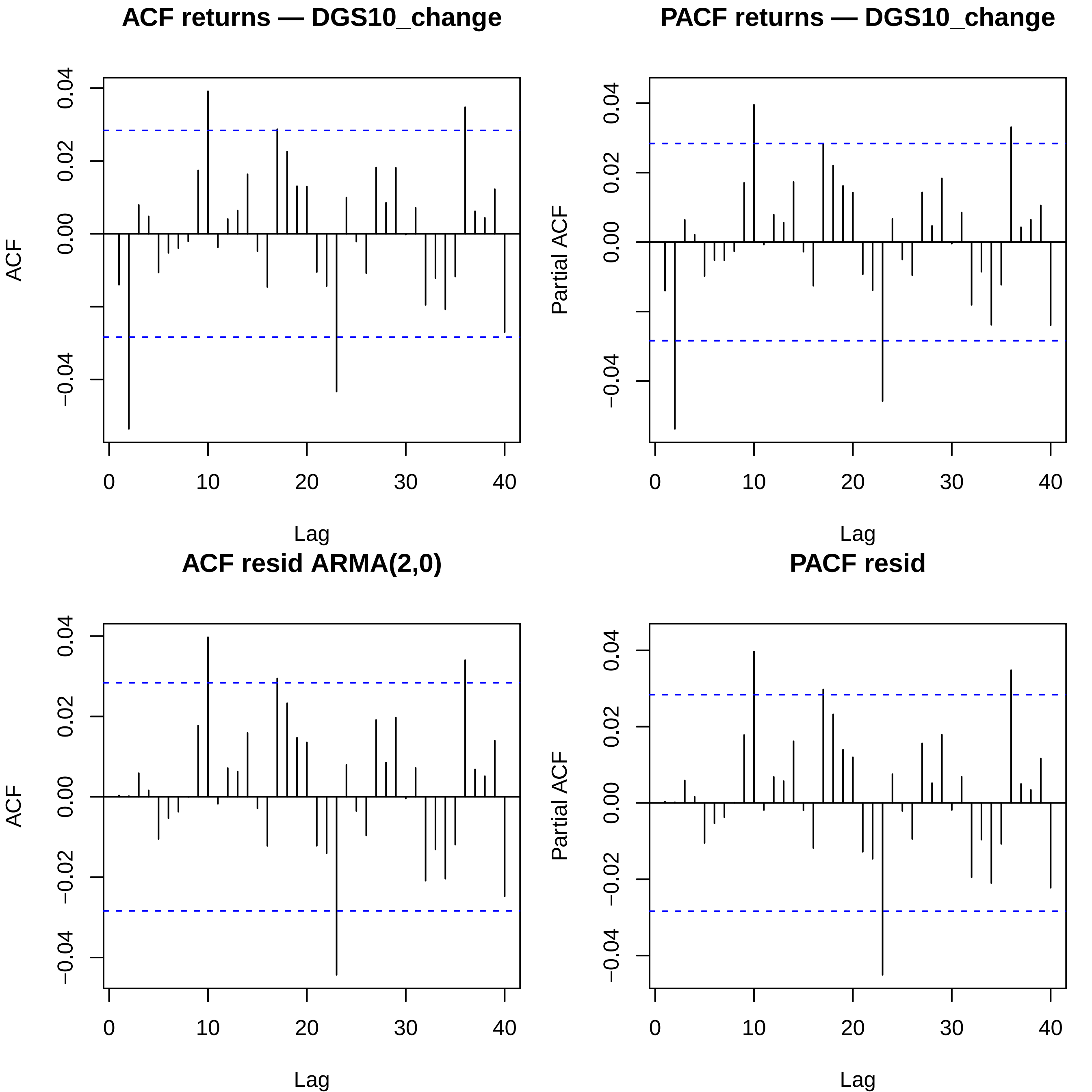
norm AIC=−6.56
std AIC=−6.63
sstd AIC=−6.64
ged AIC=−6.62
sged AIC=−6.64
nig AIC=−6.64
* jsu AIC=−6.64
empirical
theoretical

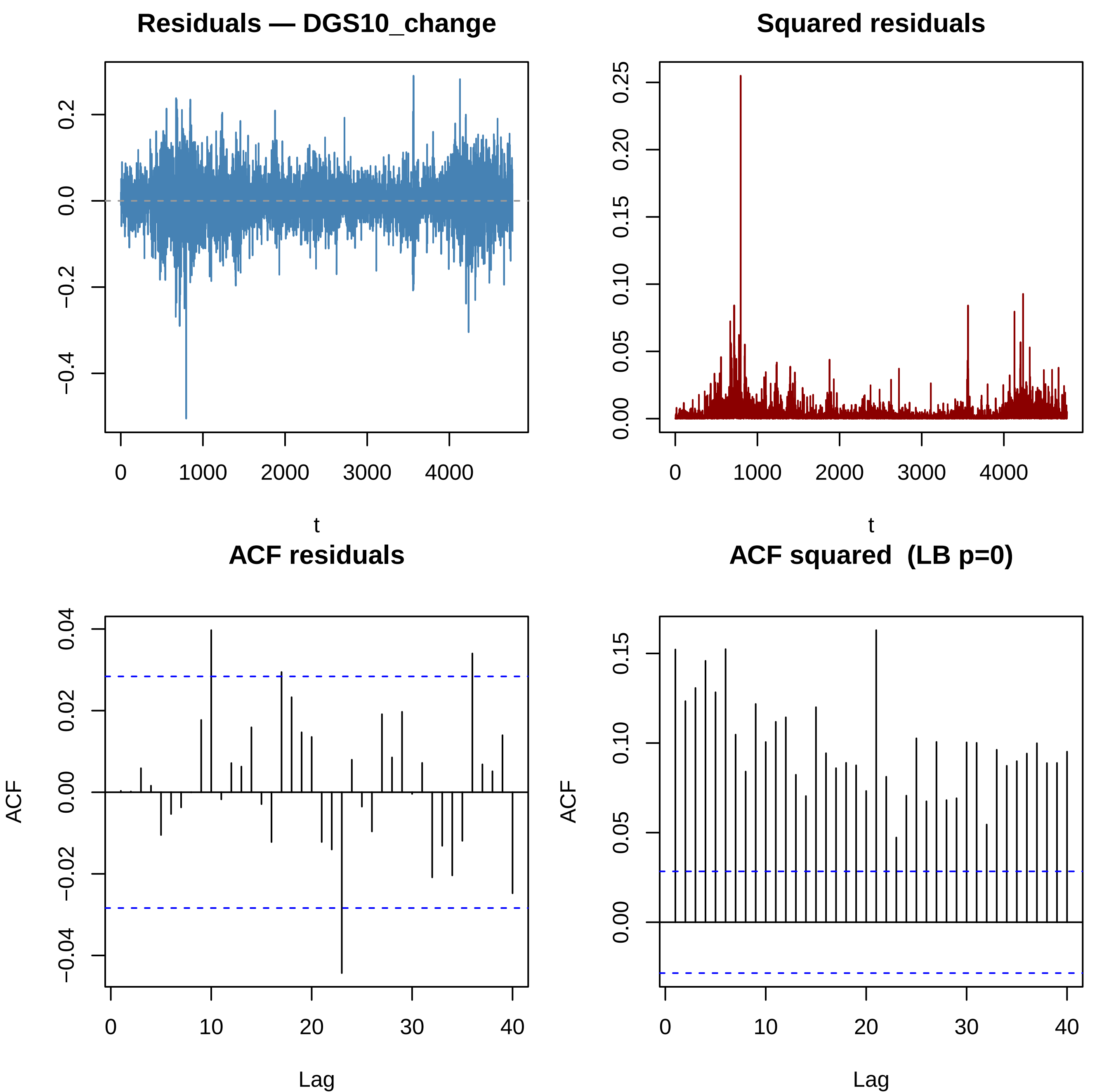

gjrGARCH(1,1)−jsu | SPY_log_ret
ret
t
Std residuals ._t
t
._t vs N(0,1)
Density
ACF . (WLB p=0.035)
ACF
Lag
ACF .² (WLB p=0.733)
ACF
Lag
Variance ratio=0.99 dev=−0.8%
s^2
t
cond s^2
emp var
model uncond


o 8 — PIT density fits to ._t — SPY_l
NO AIC=13377.9
TF AIC=13047.6
LO AIC=13068.5
SST AIC=12976
ST3 AIC=12976
* JSU AIC=12960.9
N(0,1) ref
Density
0.0
0.1
0.2
0.3
0.4
0.5
−8
−6
−4
−2
0
2
4
.


NO KS p=0
TF KS p=0.101
LO KS p=0.072
SST KS p=0.15
ST3 KS p=0.15
* JSU KS p=0.256
empirical
fitted

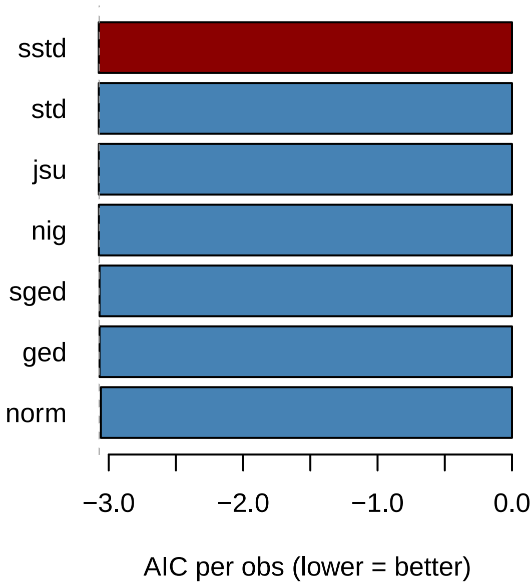

Returns — DGS10_change (ADF p=0.01)
log−ret
0.2
0.0
−0.2
−0.4
0
1000
2000
3000
4000
t
60−obs rolling mean (red) & sd (blue)
0.10
0.05
0.00
0
1000
2000
3000
4000
t

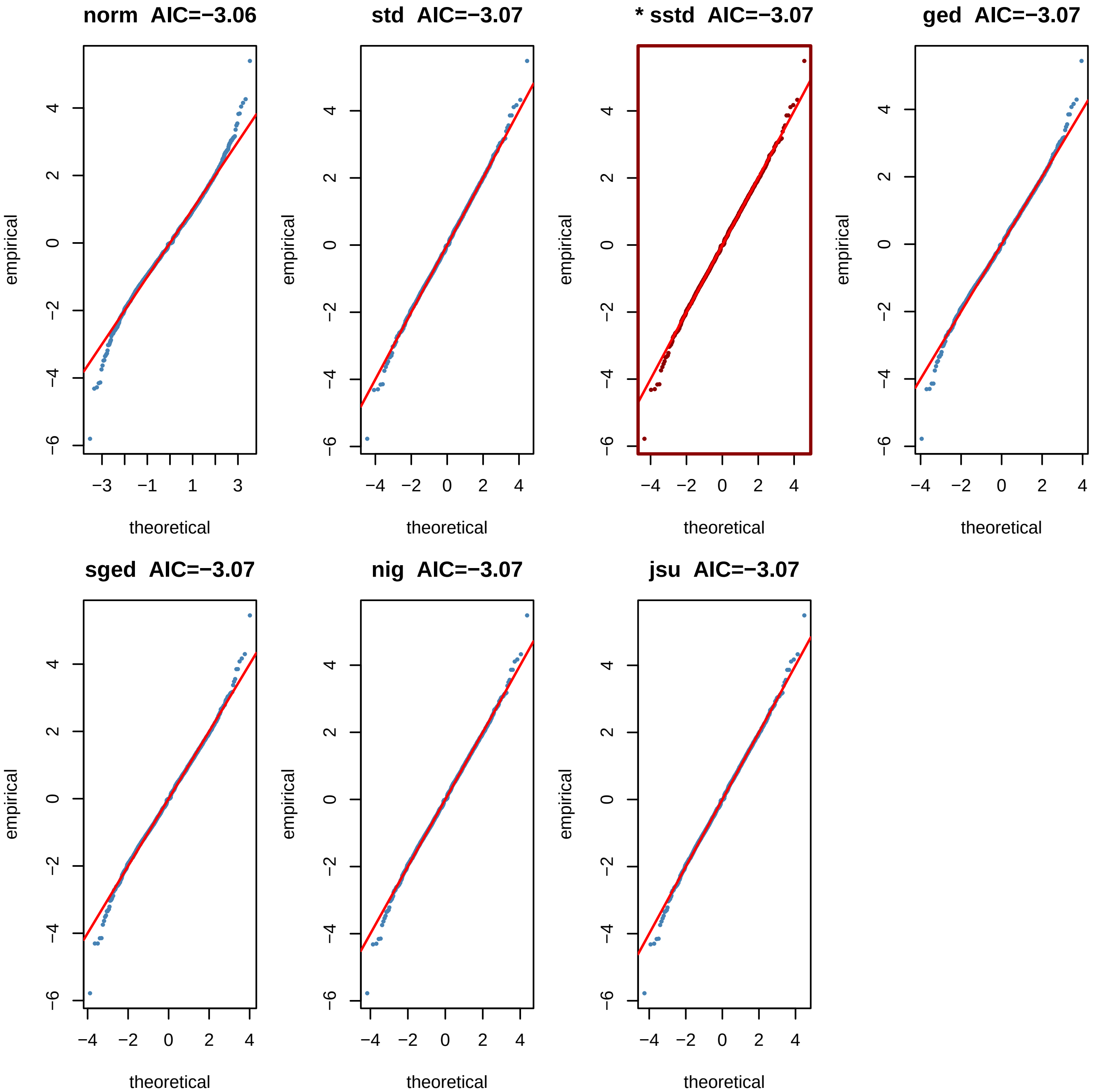

ACF returns — DGS10_change
PACF returns — DGS10_change
ACF resid ARMA(2,0)
PACF resid
ACF
Partial ACF
Lag
0.04
0.02
0.00
−0.02
−0.04
0
10
20
30
40

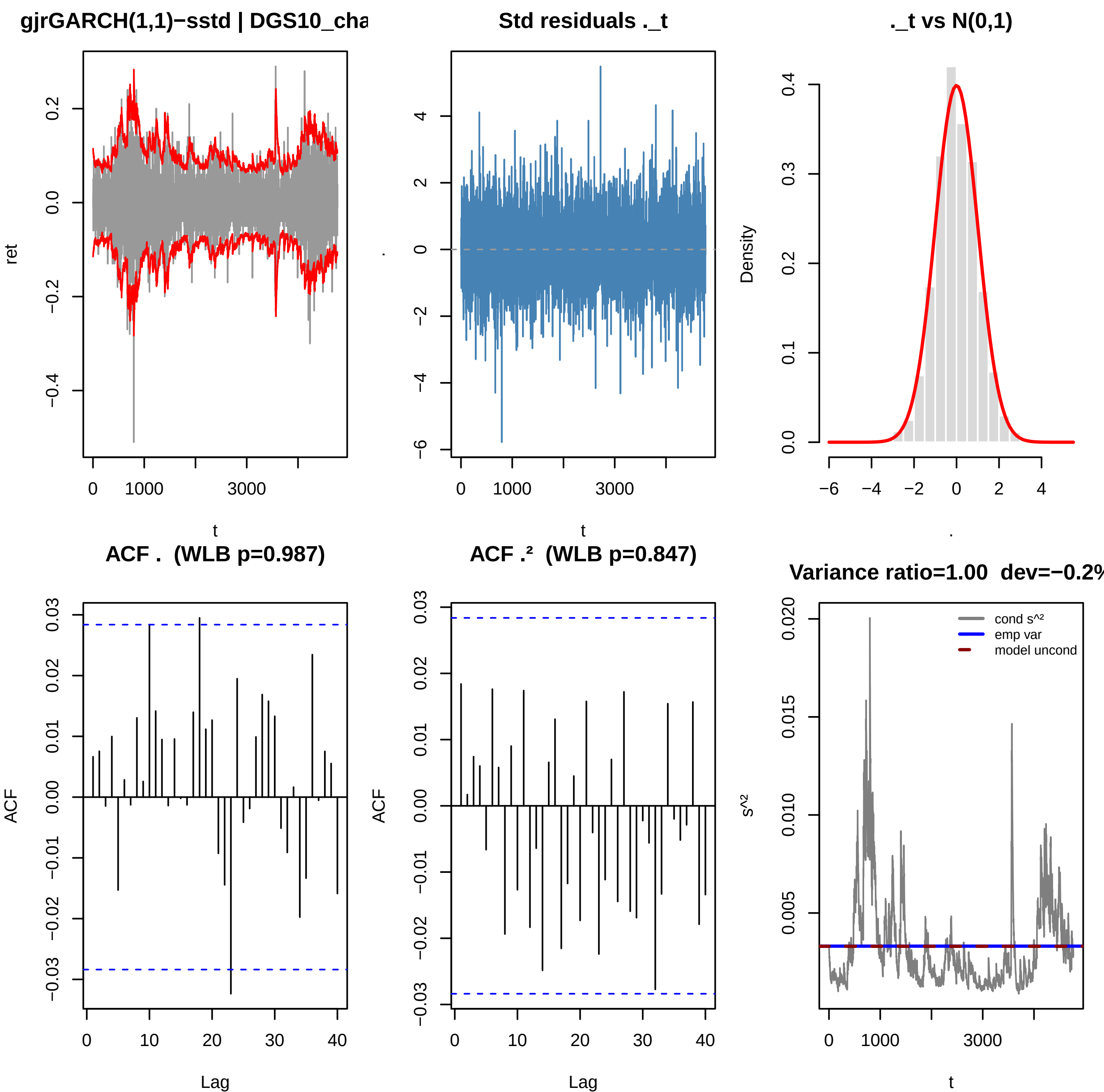


Residuals — DGS10_change
Squared residuals
ACF residuals
ACF squared (LB p=0)
t
t
Lag
Lag
ACF
ACF

Step 6 — GARCH innovation dist AIC — DGS

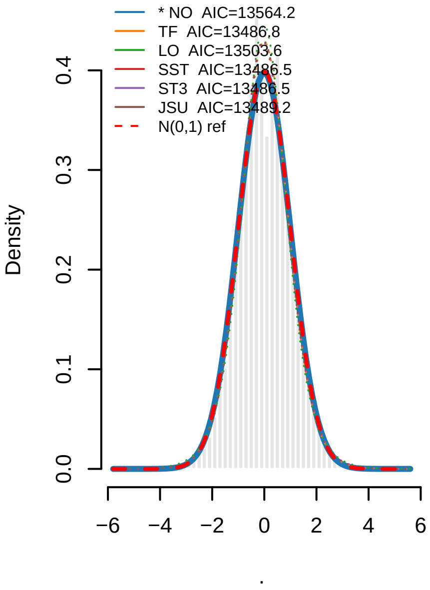


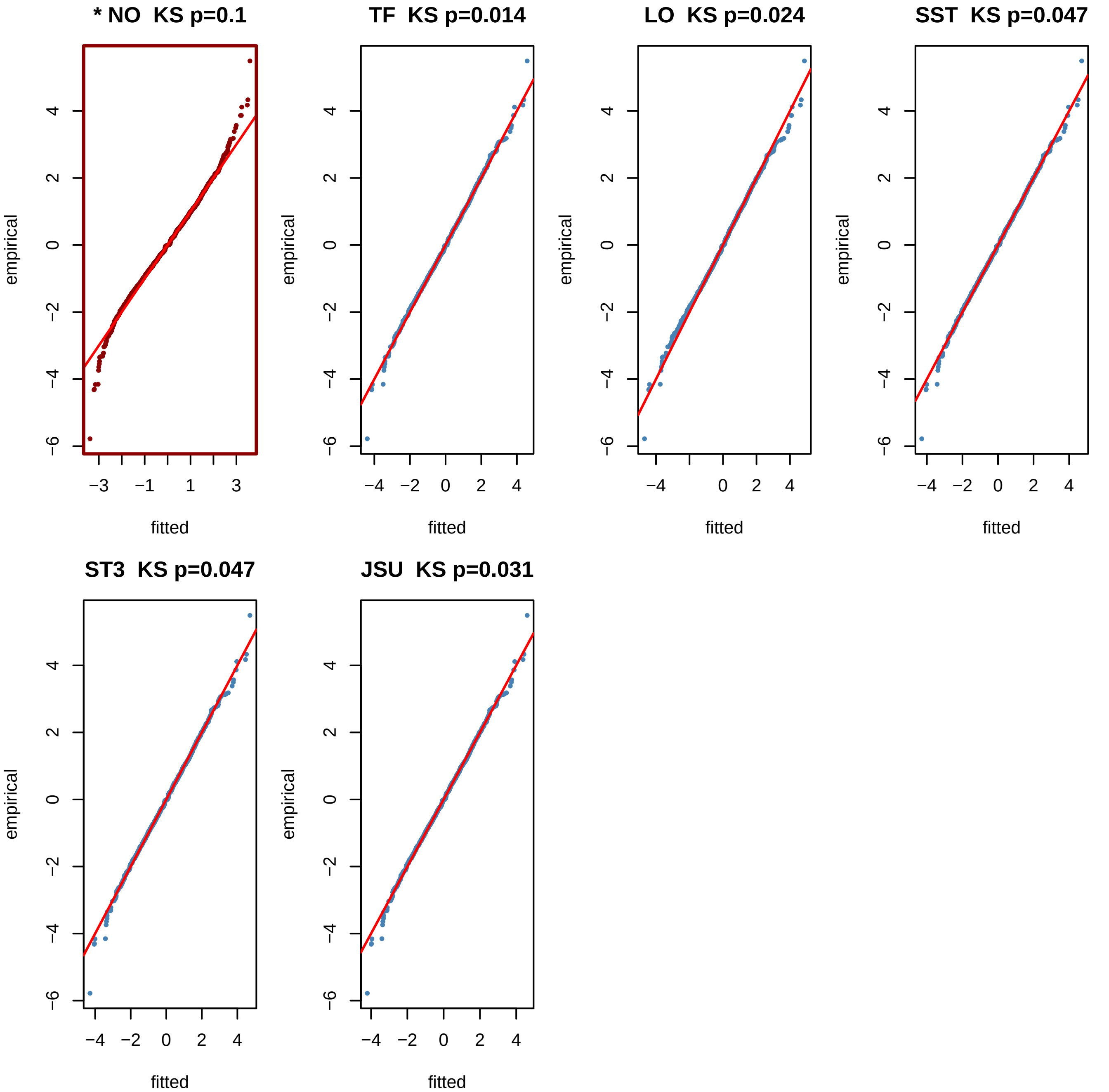

norm AIC=−3.06
std AIC=−3.07
* sstd AIC=−3.07
ged AIC=−3.07
sged AIC=−3.07
nig AIC=−3.07
jsu AIC=−3.07
empirical
theoretical
−6 −4 −2 0 2 4
−3 −1 1 3
−4 −2 0 2 4



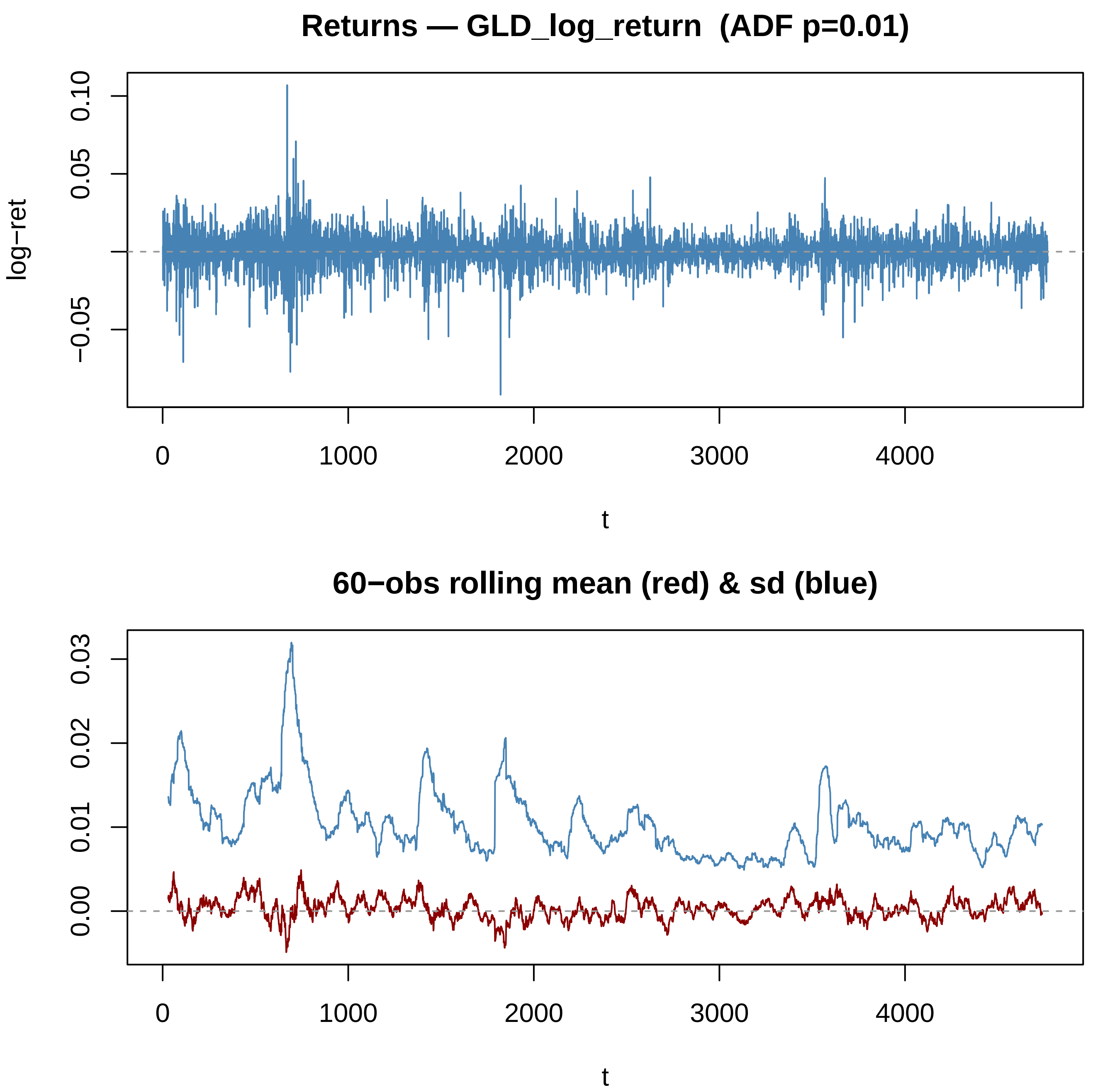
gjrGARCH(1,1)−sstd | DGS10_cha
ret
t
Std residuals ._t
t
._t vs N(0,1)
Density
ACF . (WLB p=0.987)
ACF
Lag
ACF .² (WLB p=0.847)
ACF
Lag
Variance ratio=1.00 dev=−0.2%
cond s^2
emp var
model uncond
s^2
t

p 8 — PIT density fits to ._t — DGS1

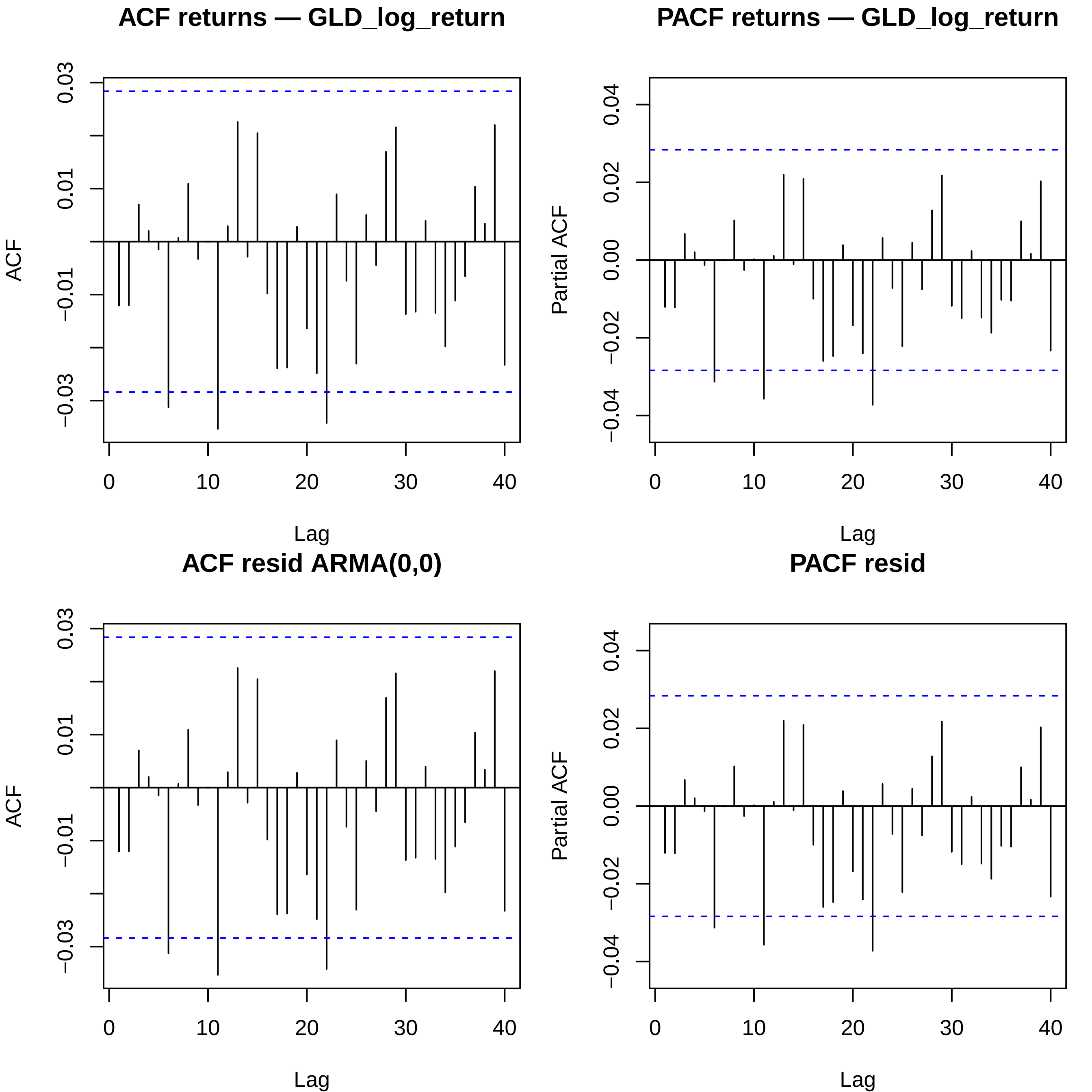


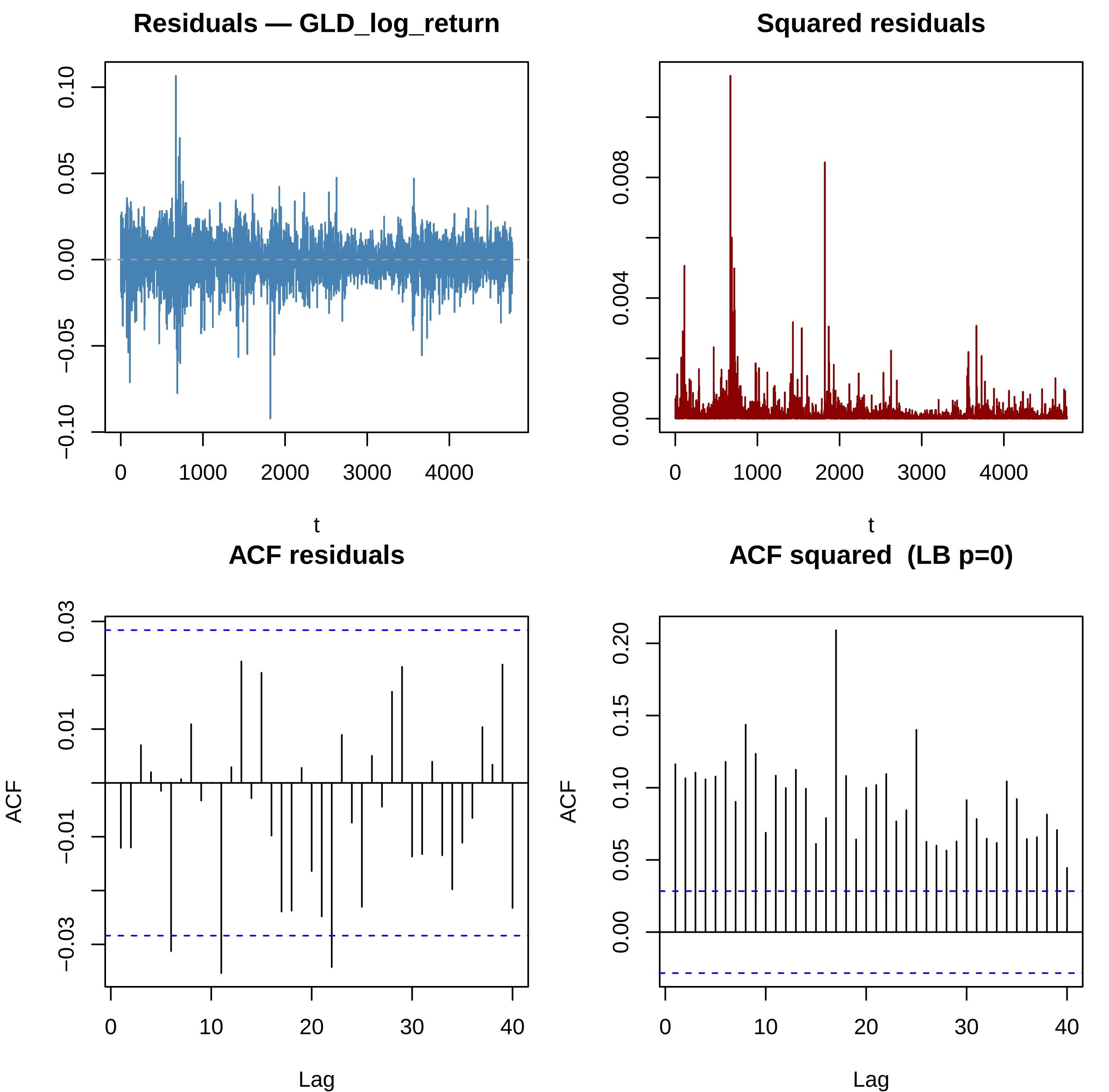


* NO KS p=0.1
TF KS p=0.014
LO KS p=0.024
SST KS p=0.047
ST3 KS p=0.047
JSU KS p=0.031
empirical
fitted

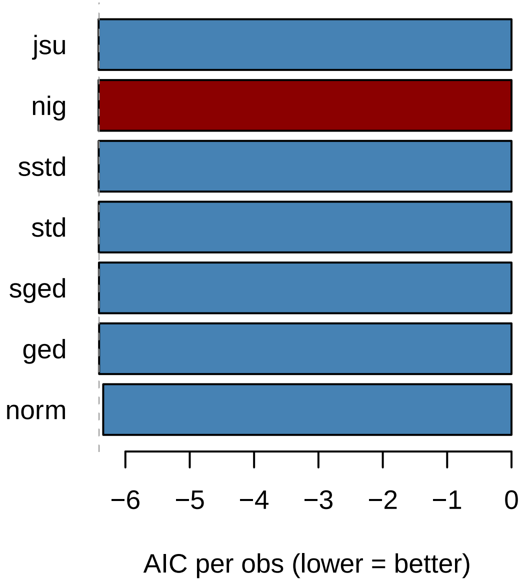


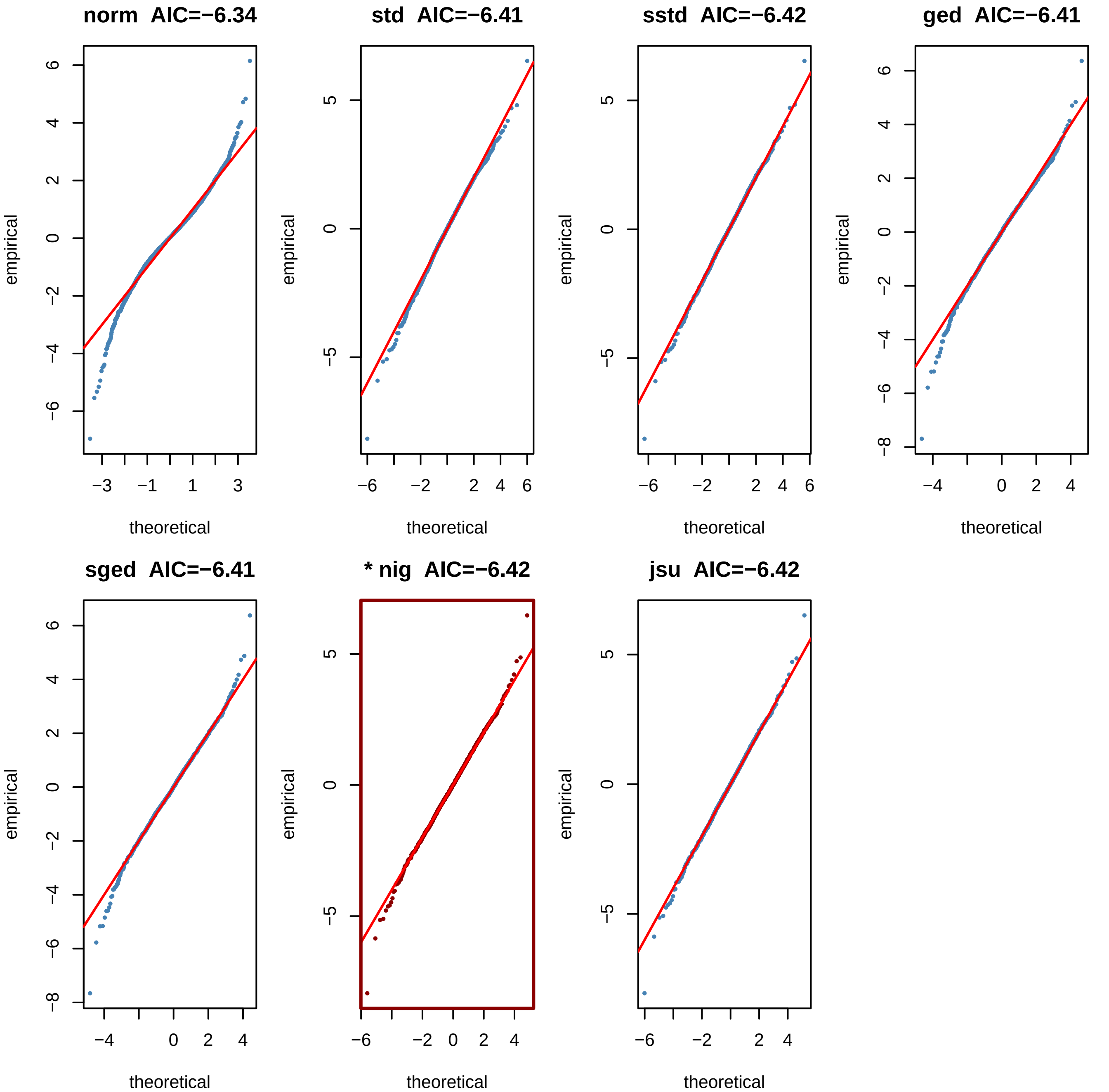
Returns — GLD_log_return (ADF p=0.01)
log−ret
0.10
0.05
−0.05
0
1000
2000
3000
4000
t
60−obs rolling mean (red) & sd (blue)
0.03
0.02
0.01
0.00
0
1000
2000
3000
4000
t

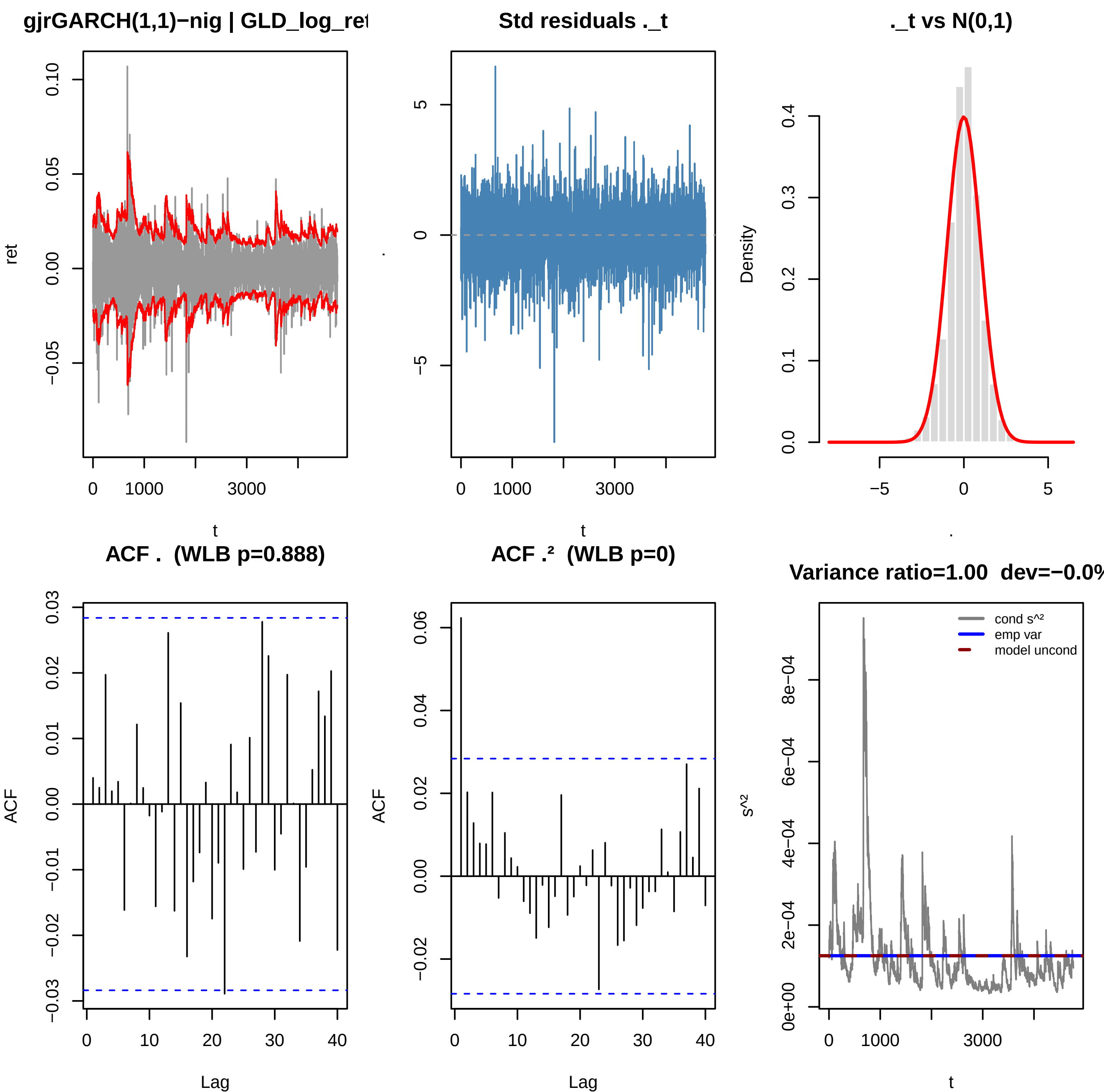

ACF returns — GLD_log_return
PACF returns — GLD_log_return
ACF resid ARMA(0,0)
PACF resid
ACF
Partial ACF
Lag
0.03
0.01
−0.01
−0.03
0.04
0.02
0.00
−0.02
−0.04
0
10
20
30
40


Residuals — GLD_log_return
Squared residuals
ACF residuals
ACF squared (LB p=0)
t
t
Lag
Lag
ACF
ACF
0.10
0.05
0.00
−0.05
−0.10
0.008
0.004
0.000
0.03
0.01
−0.01
−0.03
0.20
0.15
0.10
0.05
0.00
0
1000
2000
3000
4000
0
10
20
30
40

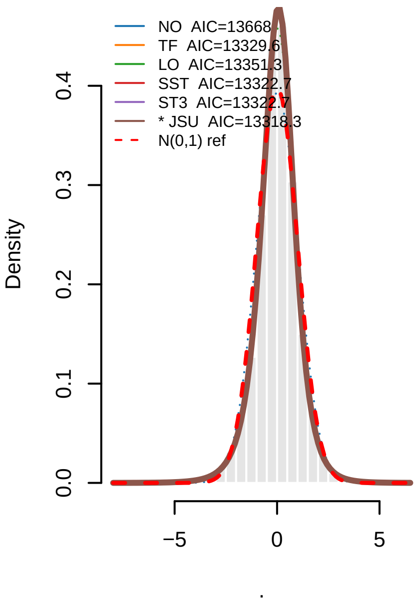
Step 6 — GARCH innovation dist AIC — GLD

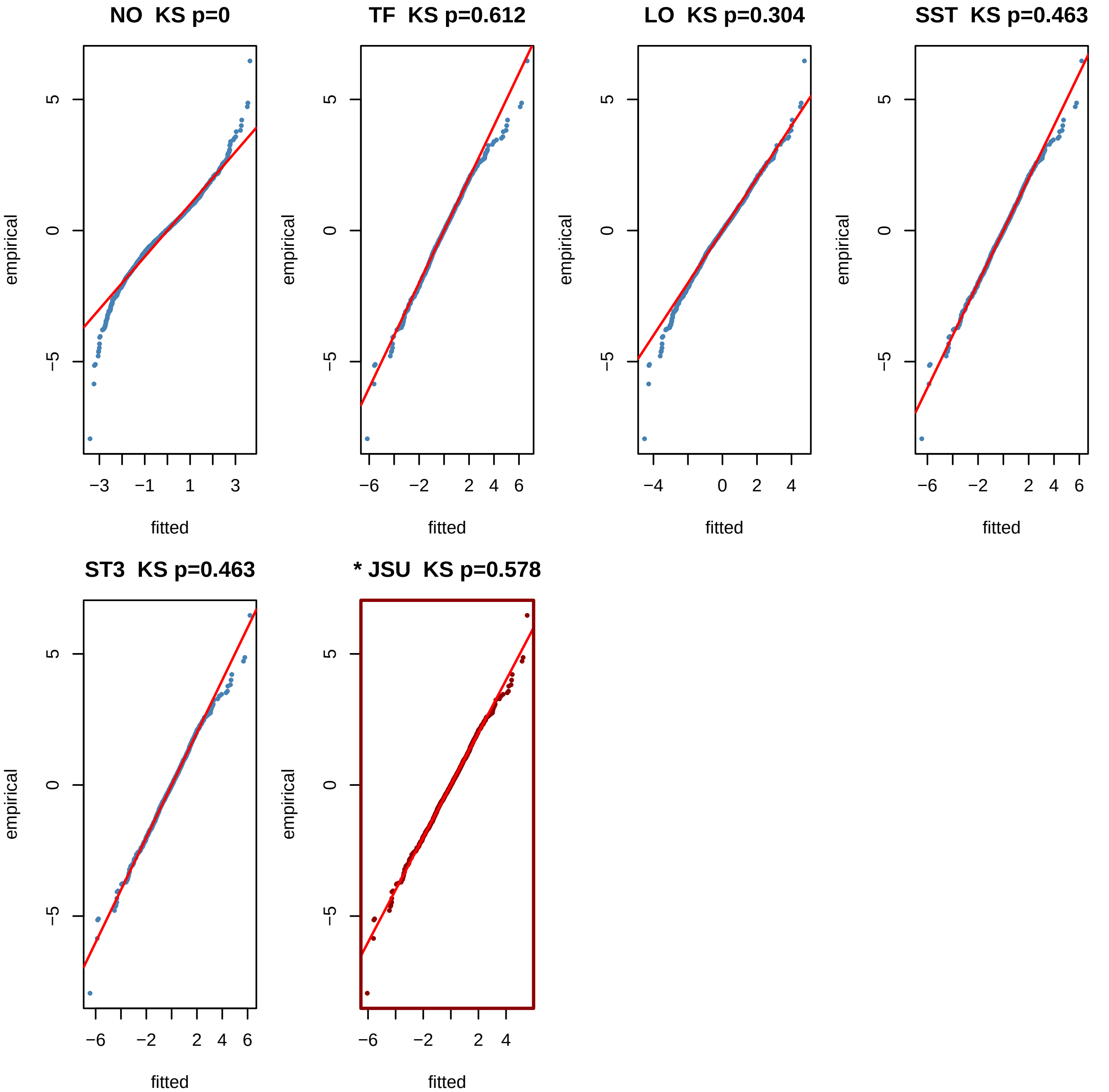



norm AIC=−6.34
std AIC=−6.41
sstd AIC=−6.42
ged AIC=−6.41
sged AIC=−6.41
* nig AIC=−6.42
jsu AIC=−6.42
empirical
theoretical

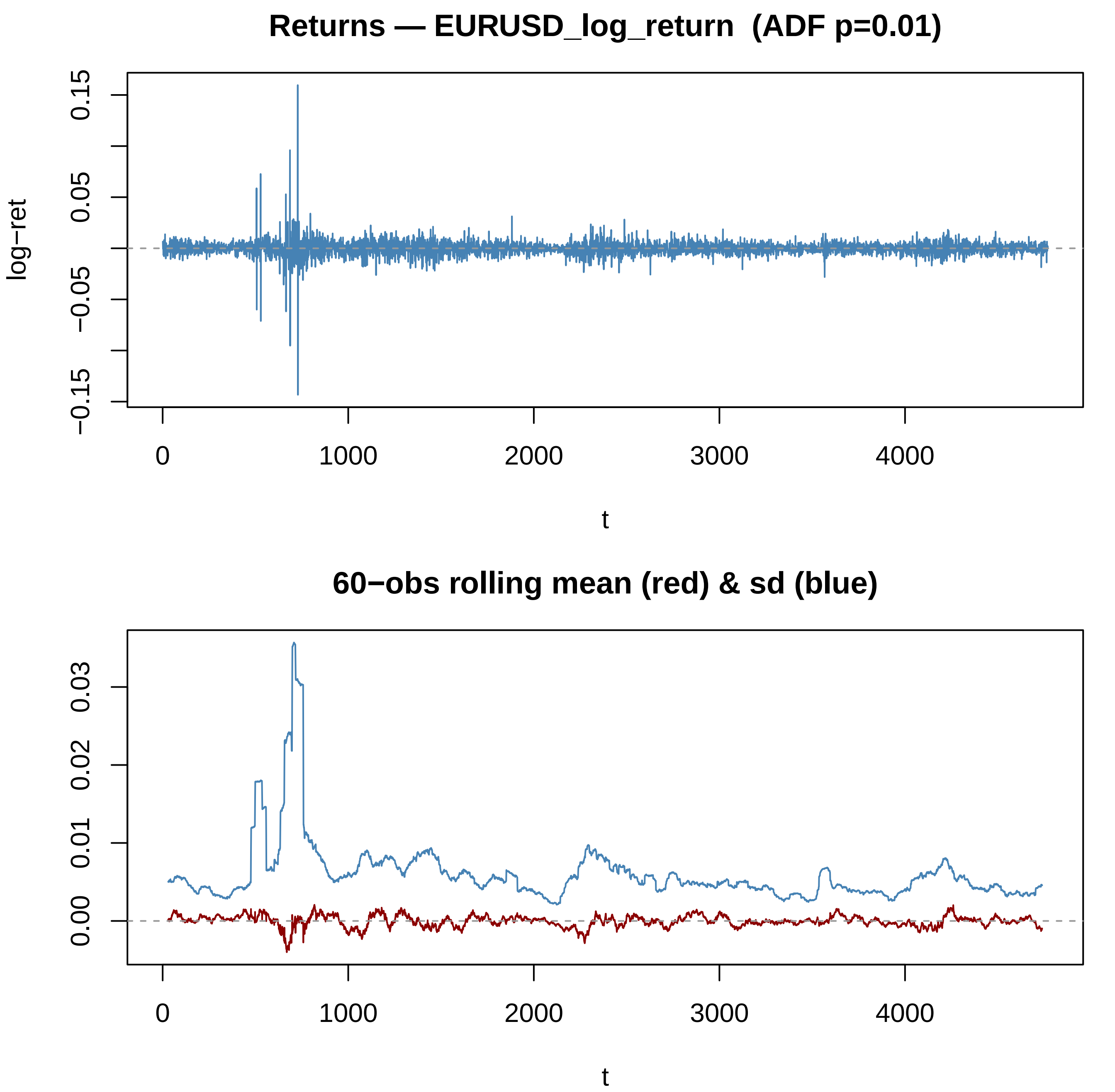

gjrGARCH(1,1)−nig | GLD_log_ret
ret
t
Std residuals ._t
t
._t vs N(0,1)
Density
ACF . (WLB p=0.888)
ACF
Lag
ACF .² (WLB p=0)
ACF
Lag
Variance ratio=1.00 dev=−0.0%
cond s^2
emp var
model uncond
s^2
t

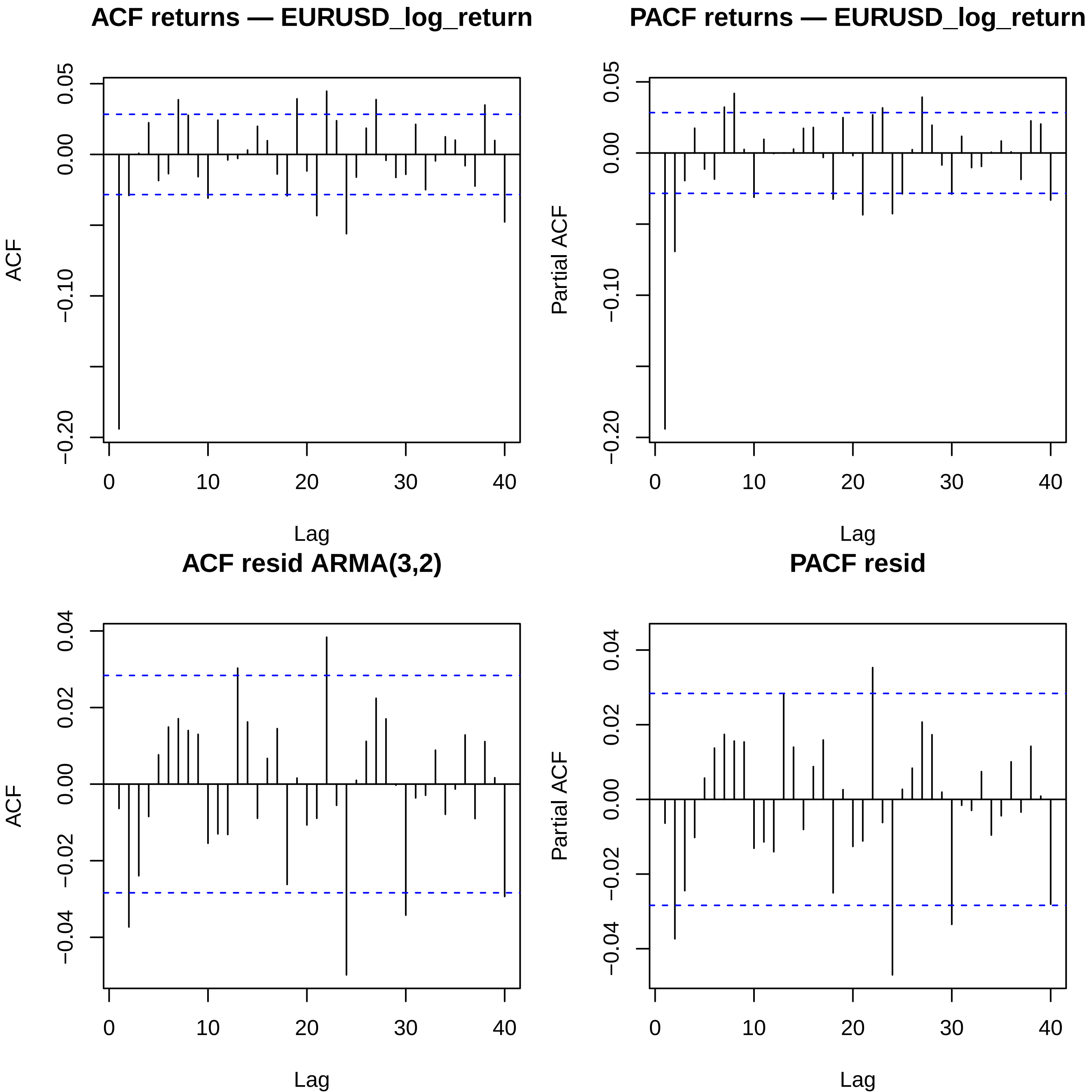
o 8 — PIT density fits to ._t — GLD_l

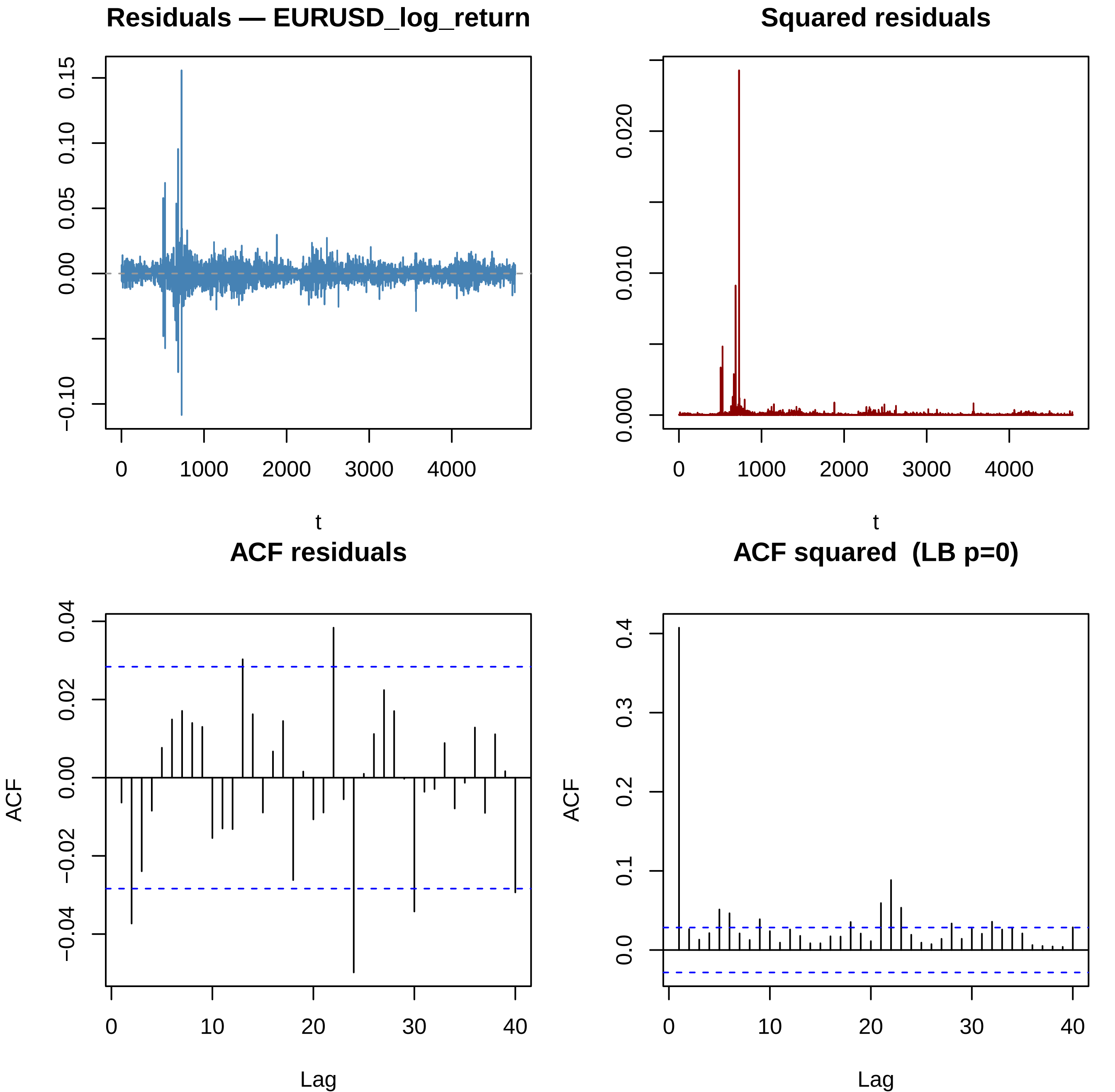




NO KS p=0
TF KS p=0.612
LO KS p=0.304
SST KS p=0.463
ST3 KS p=0.463
* JSU KS p=0.578
empirical
fitted

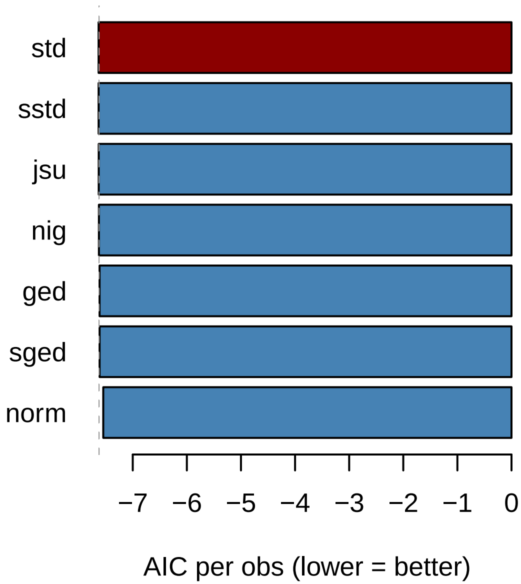

Returns — EURUSD_log_return (ADF p=0.01)
log−ret
0.15
0.05
−0.05
−0.15
0
1000
2000
3000
4000
t
60−obs rolling mean (red) & sd (blue)
0.03
0.02
0.01
0.00
0
1000
2000
3000
4000
t

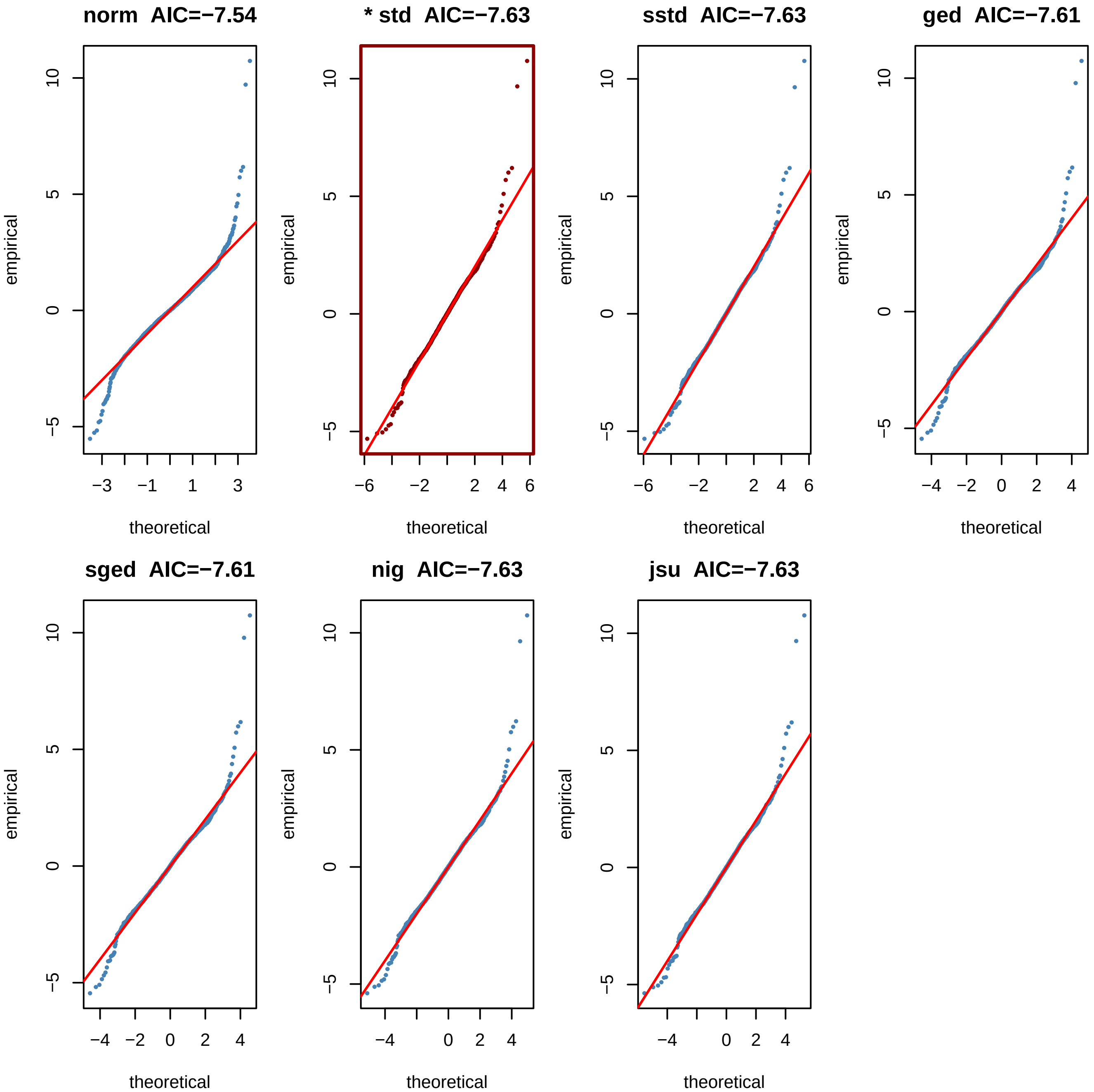

ACF returns — EURUSD_log_return
PACF returns — EURUSD_log_return
ACF resid ARMA(3,2)
PACF resid
ACF
Partial ACF
Lag
0.05
0.00
−0.10
−0.20
0.04
0.02
−0.02
−0.04
0
10
20
30
40



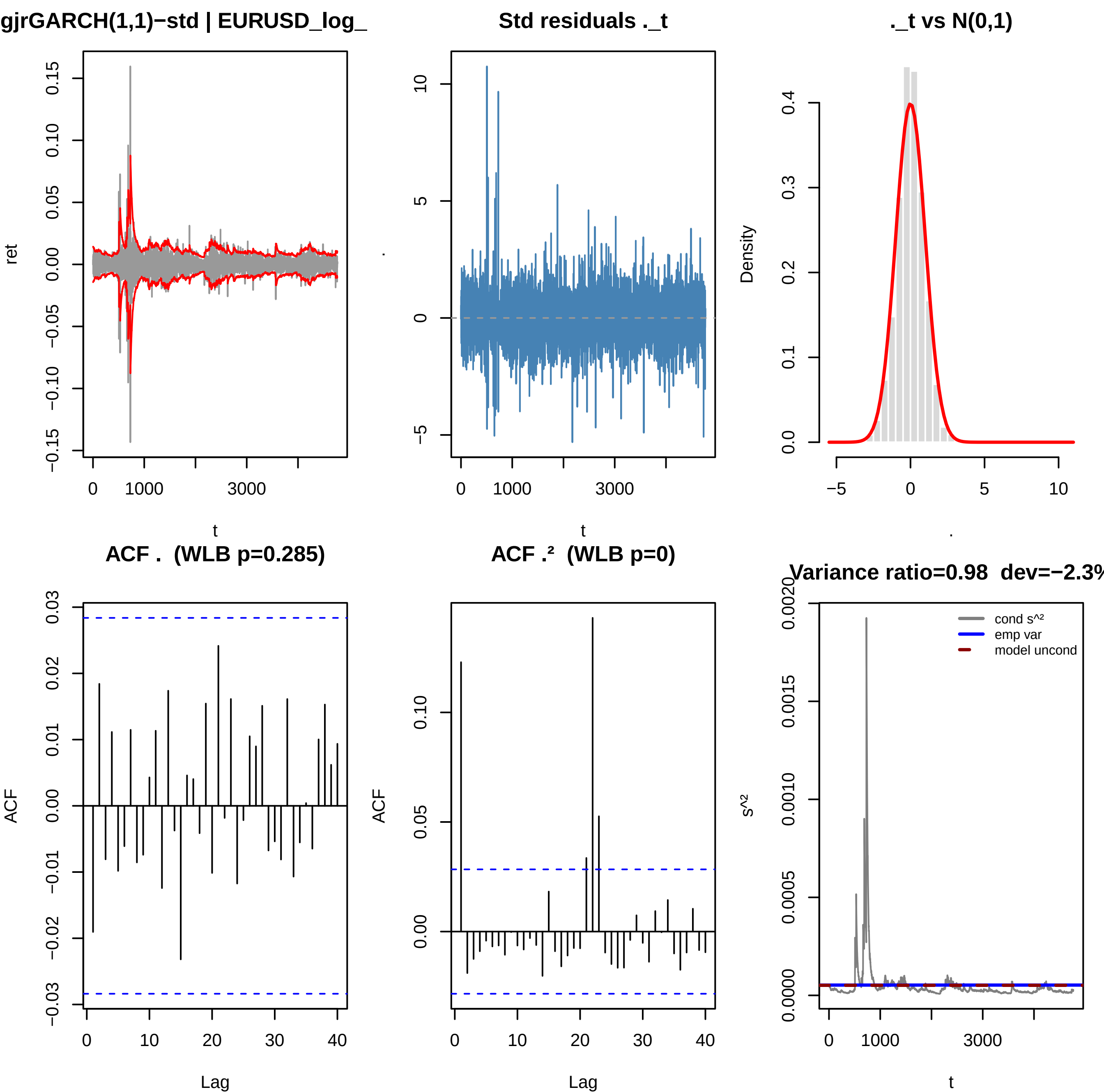
Residuals — EURUSD_log_return
Squared residuals
ACF residuals
ACF squared (LB p=0)
t
Lag
ACF
0.15
0.10
0.05
0.00
−0.10
0.020
0.010
0.000
0.04
0.02
−0.02
−0.04
0.4
0.3
0.2
0.1
0.0
0
1000
2000
3000
4000
10
20
30
40

tep 6 — GARCH innovation dist AIC — EURU

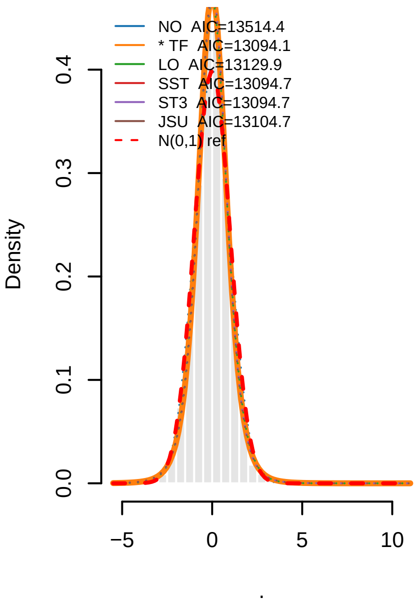


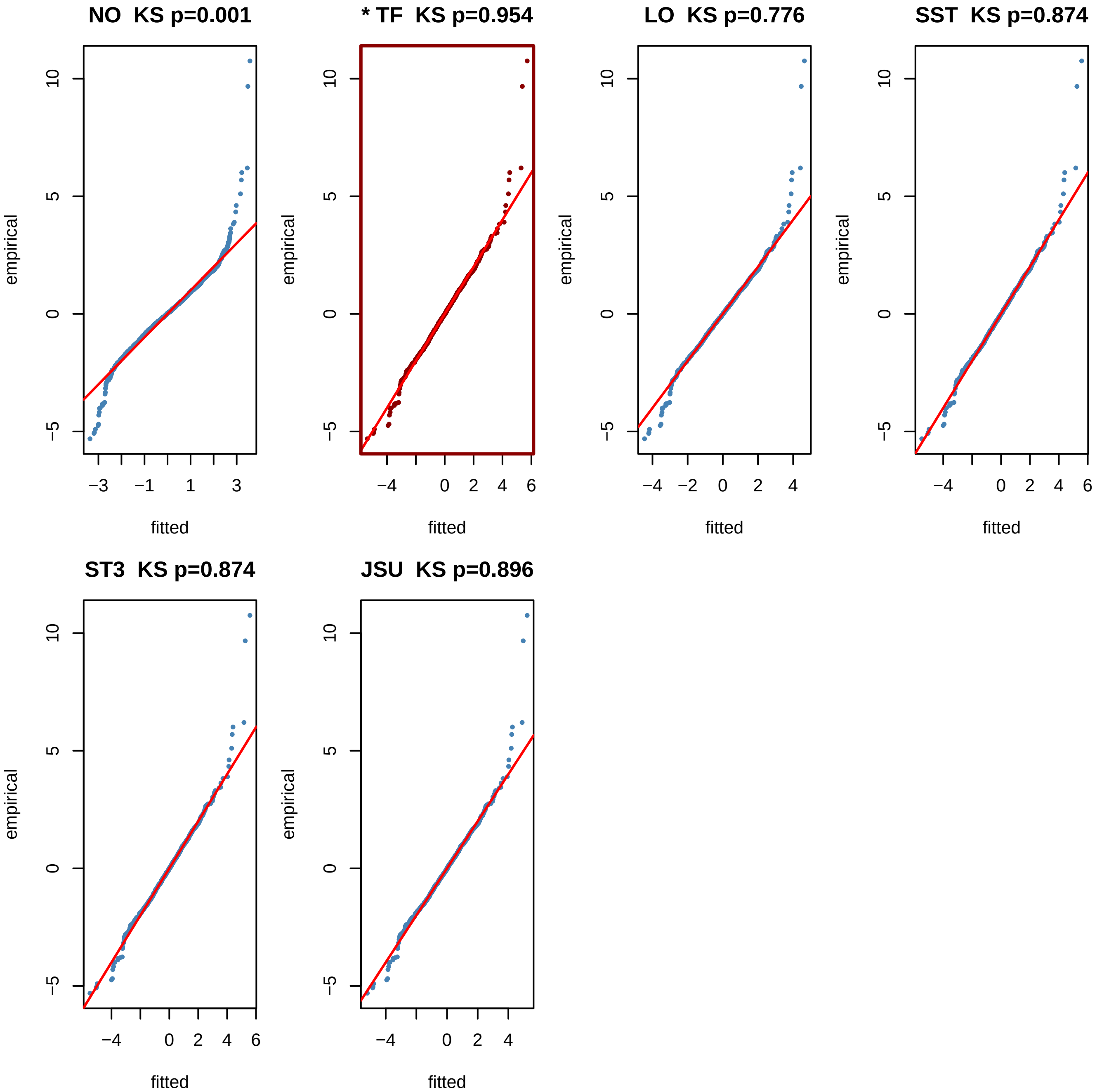


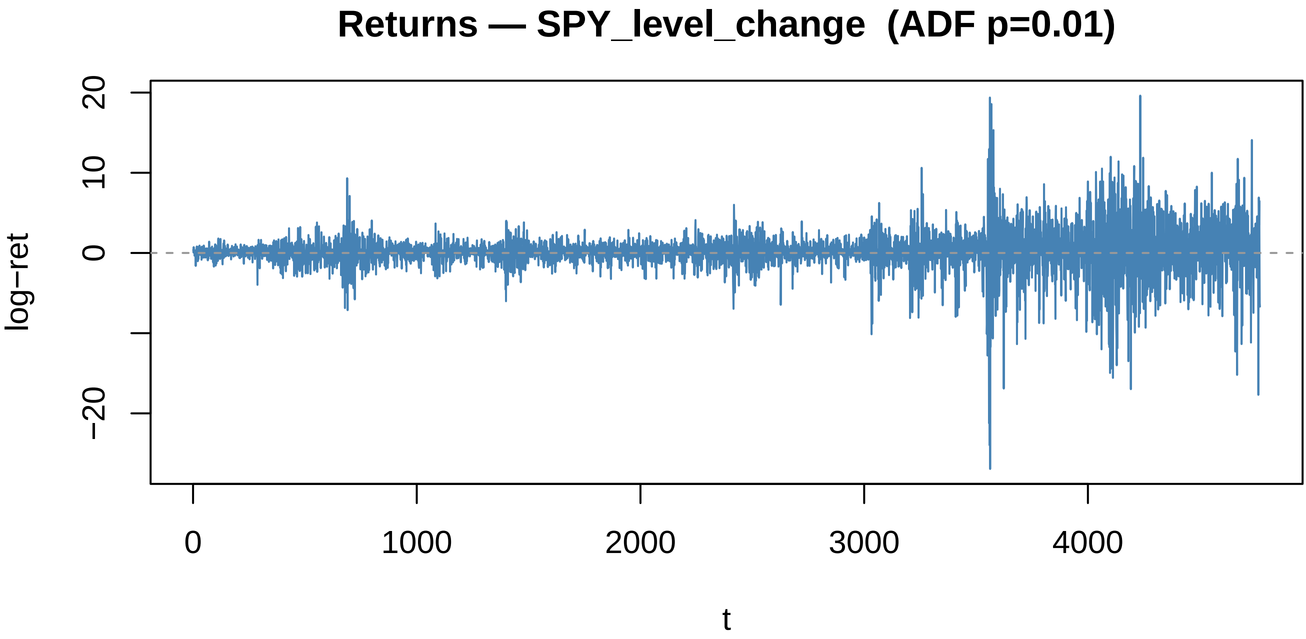
norm AIC=−7.54
* std AIC=−7.63
sstd AIC=−7.63
ged AIC=−7.61
sged AIC=−7.61
nig AIC=−7.63
jsu AIC=−7.63
empirical
theoretical

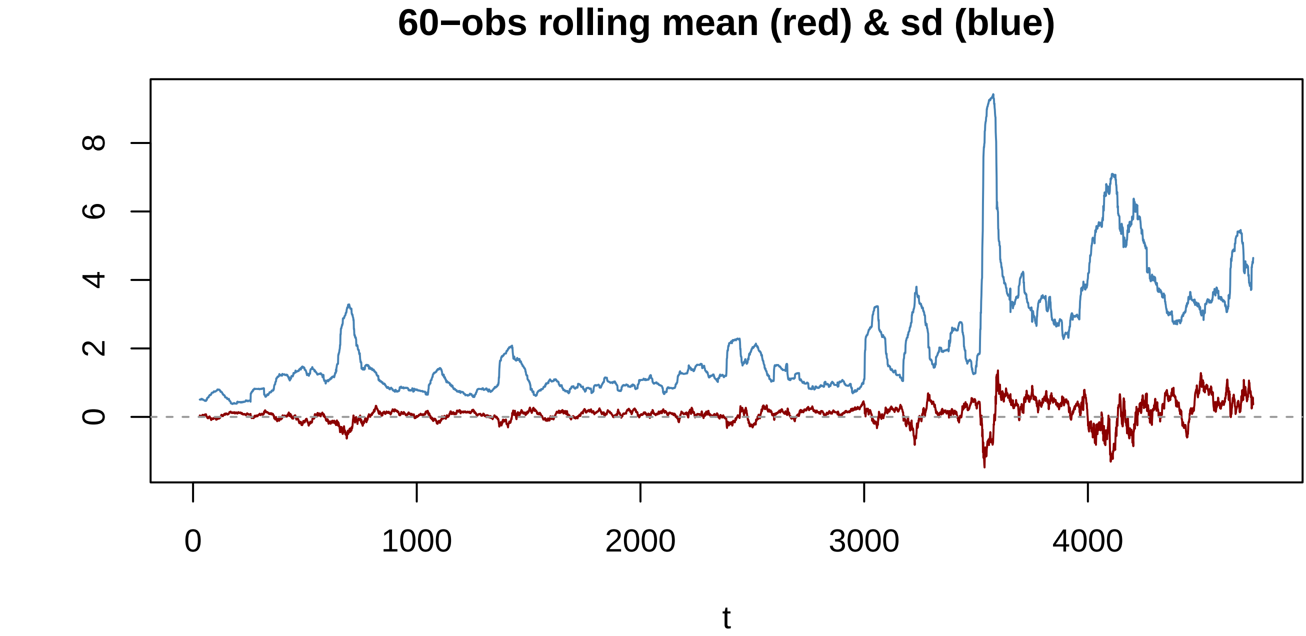

gjrGARCH(1,1)−std | EURUSD_log_
ret
t
Std residuals ._t
t
._t vs N(0,1)
Density
ACF . (WLB p=0.285)
ACF
Lag
ACF .² (WLB p=0)
ACF
Lag
Variance ratio=0.98 dev=−2.3%
cond s^2
emp var
model uncond
s^2
t

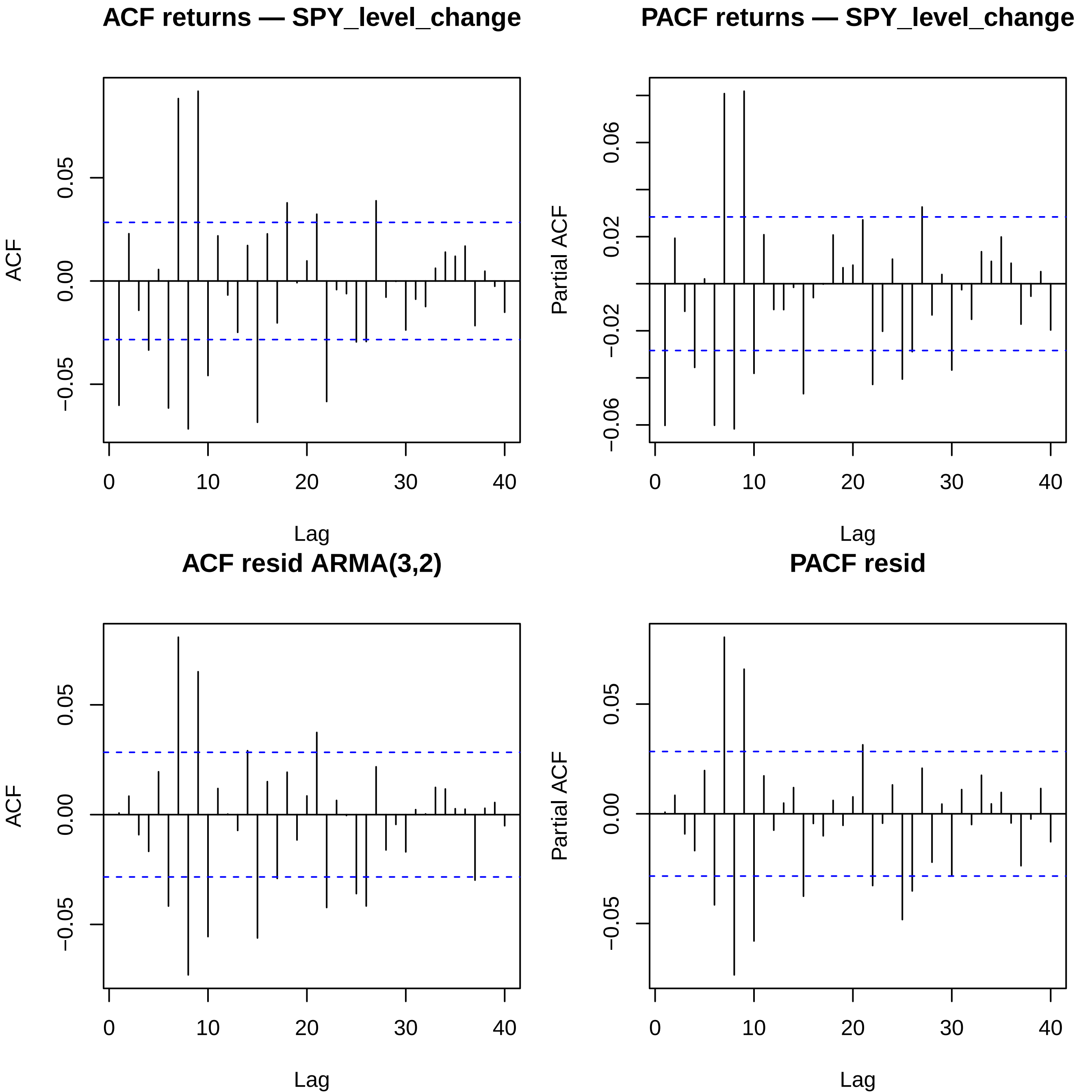
— PIT density fits to ._t — EURUSD

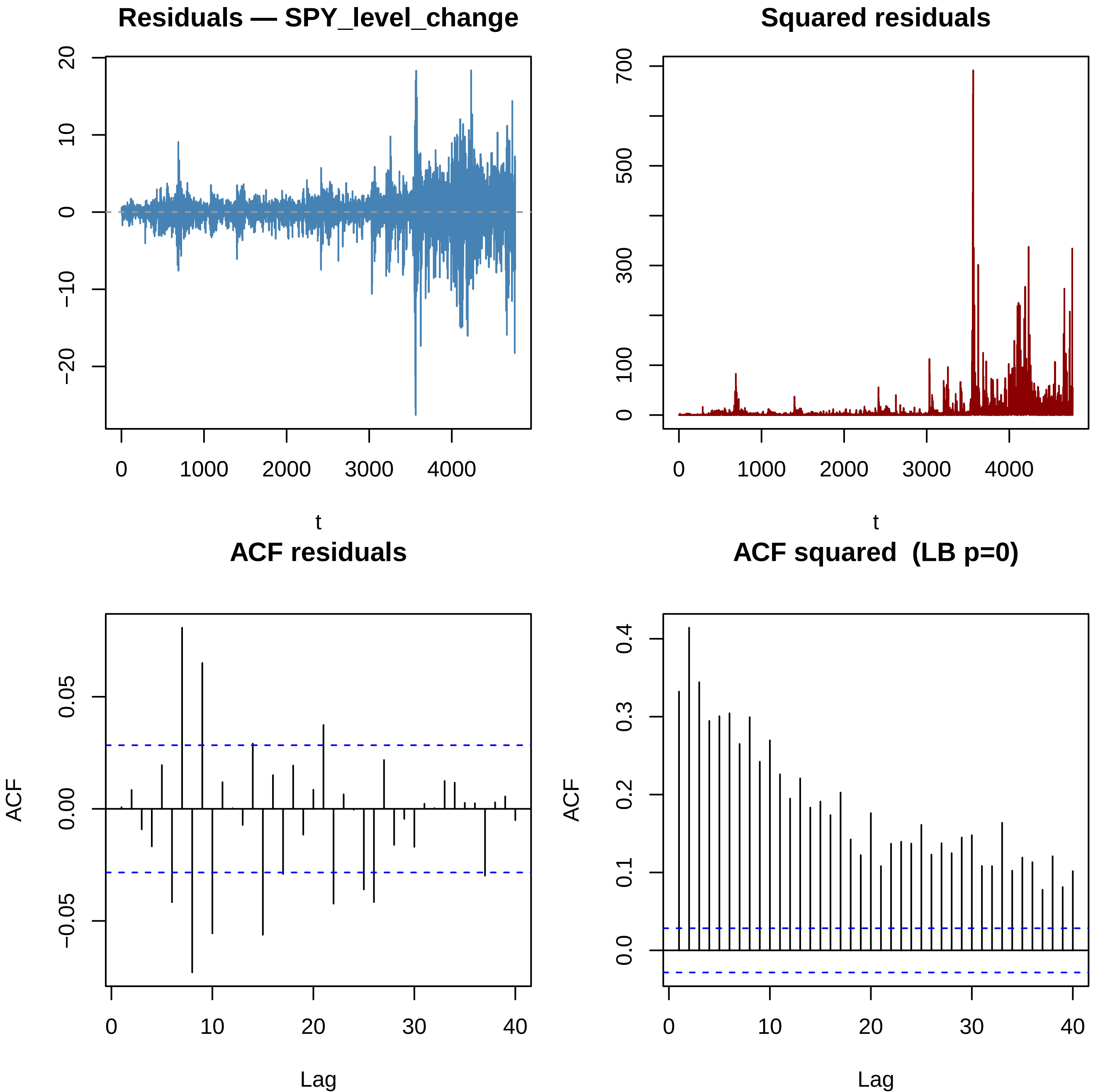



NO KS p=0.001
* TF KS p=0.954
LO KS p=0.776
SST KS p=0.874
ST3 KS p=0.874
JSU KS p=0.896
empirical
fitted

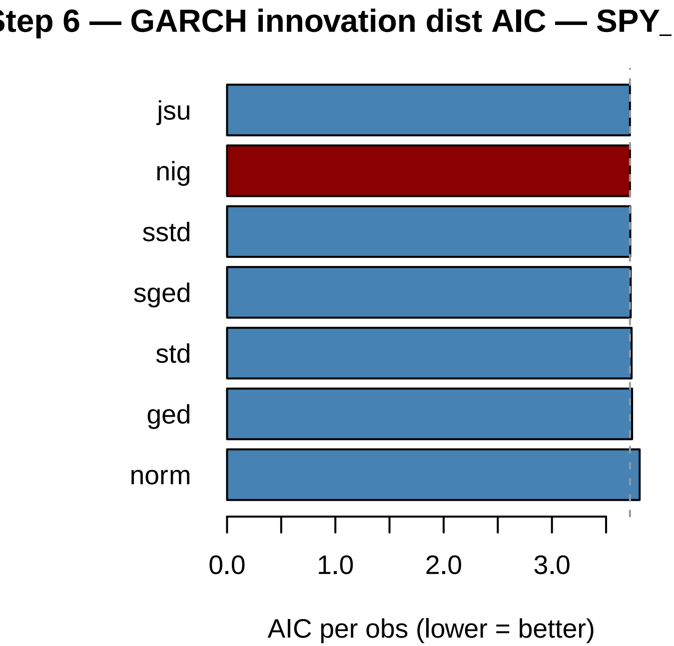

Returns — SPY_level_change (ADF p=0.01)
log−ret
20
10
0
−20
0
1000
2000
3000
4000
t

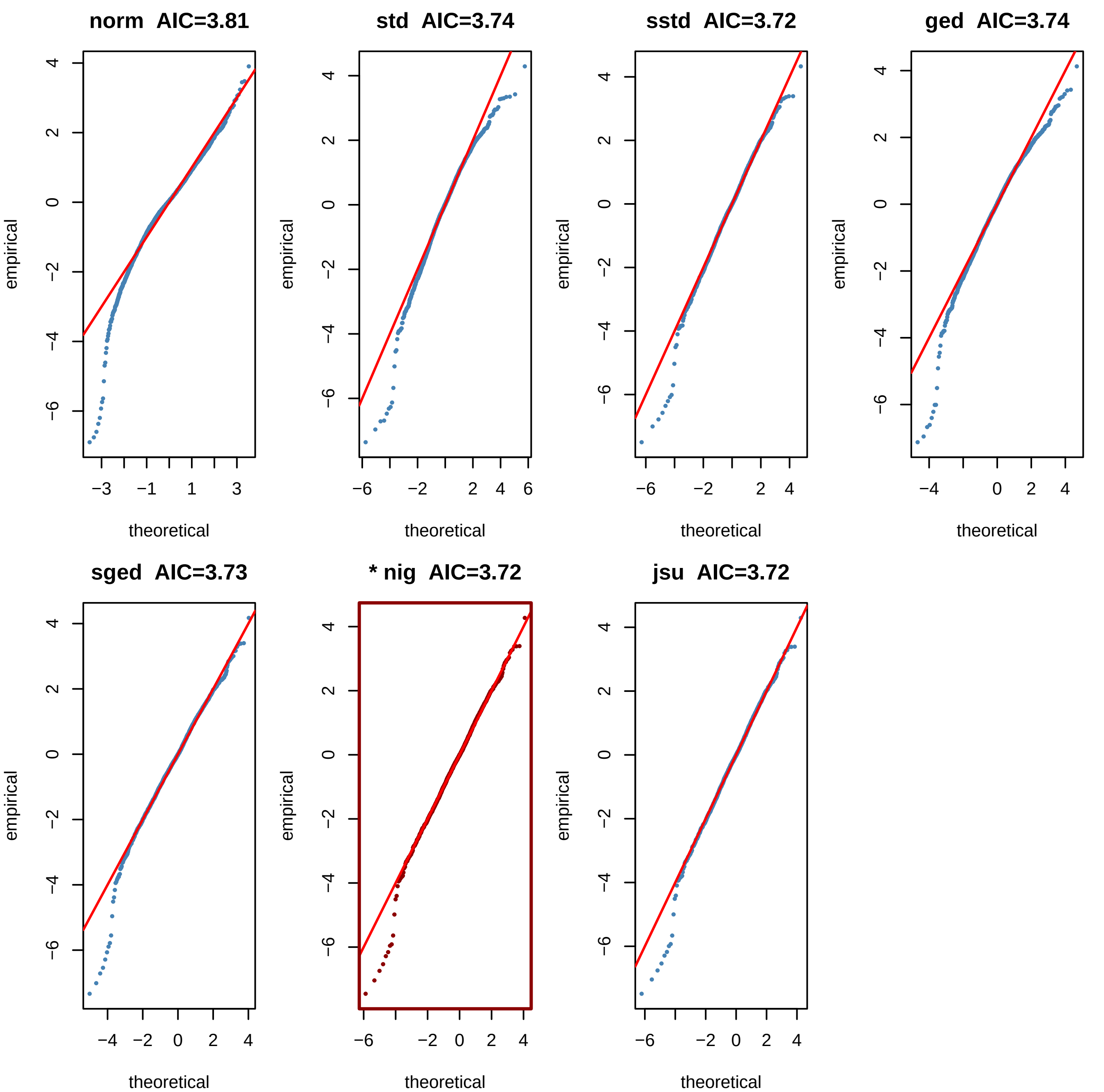

60−obs rolling mean (red) & sd (blue)
8
6
4
2
0
0
1000
2000
3000
4000
t

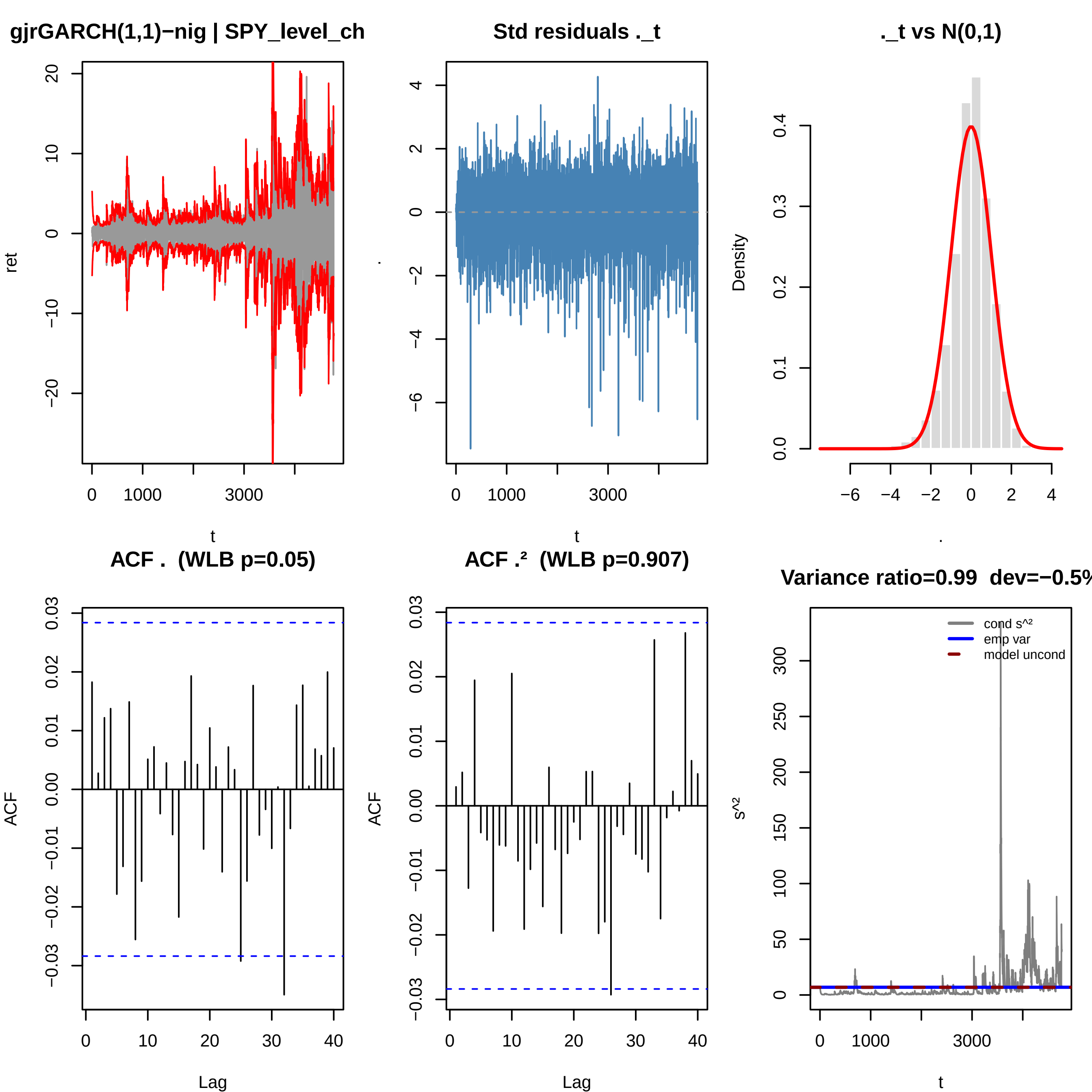

ACF returns — SPY_level_change
PACF returns — SPY_level_change
ACF resid ARMA(3,2)
PACF resid
ACF
Partial ACF
Lag
0.05
0.00
−0.05
0.06
0.02
−0.02
−0.06
0
10
20
30
40



Residuals — SPY_level_change
Squared residuals
ACF residuals
ACF squared (LB p=0)
t
Lag
ACF

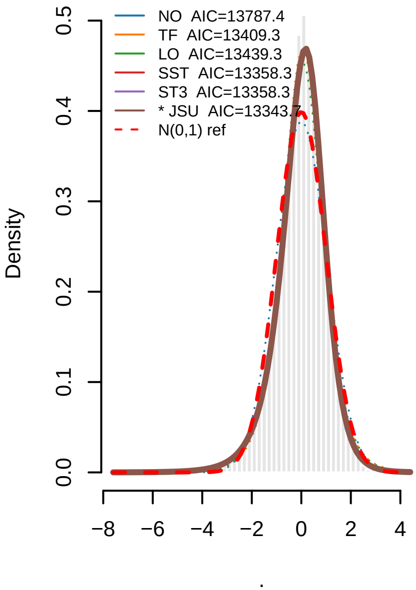

Step 6 — GARCH innovation dist AIC — SPY_
jsu
nig
sstd
sged
std
ged
norm
0.0
1.0
2.0
3.0
AIC per obs (lower = better)

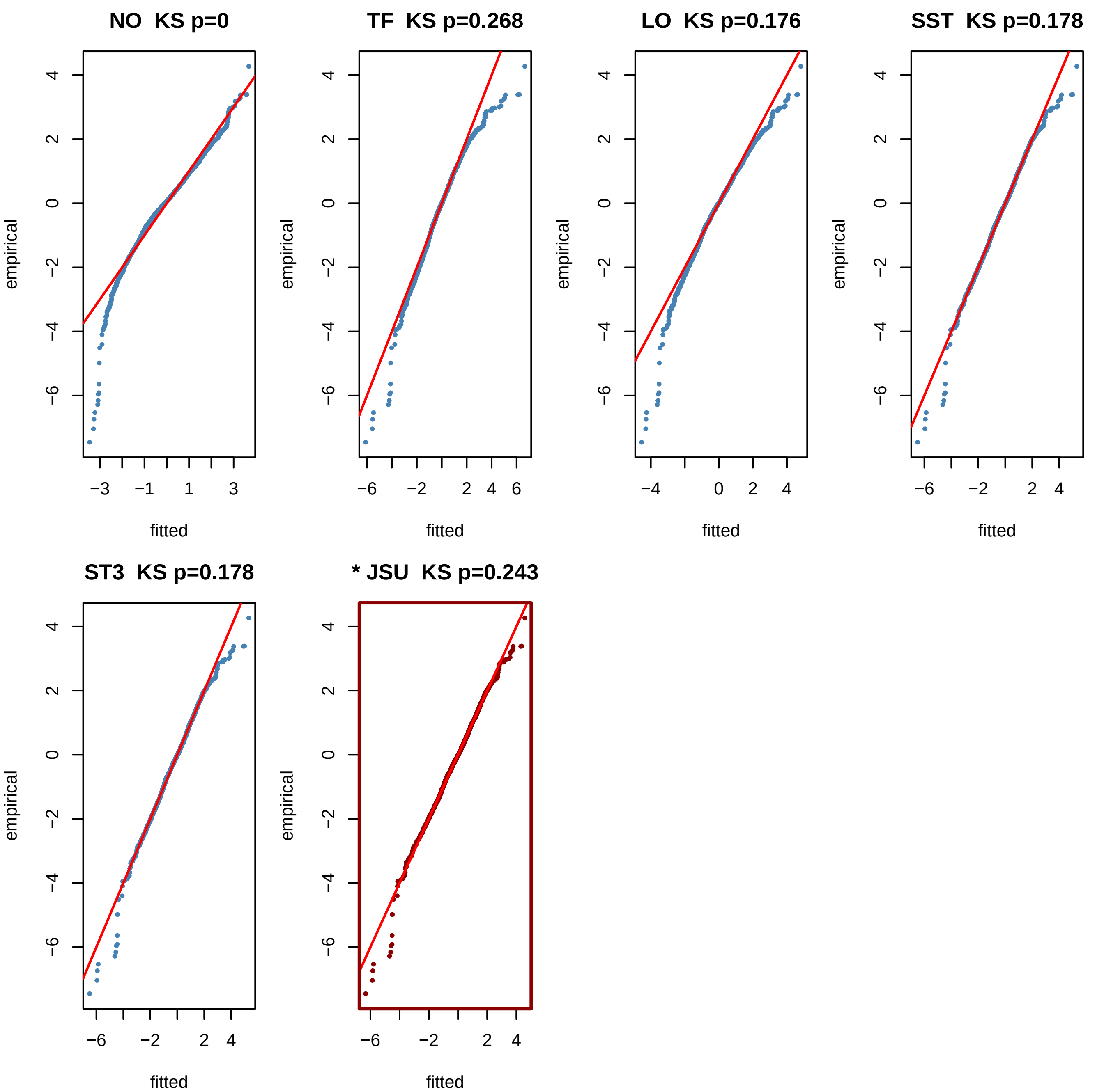

norm AIC=3.81
std AIC=3.74
sstd AIC=3.72
ged AIC=3.74
sged AIC=3.73
* nig AIC=3.72
jsu AIC=3.72
empirical
theoretical

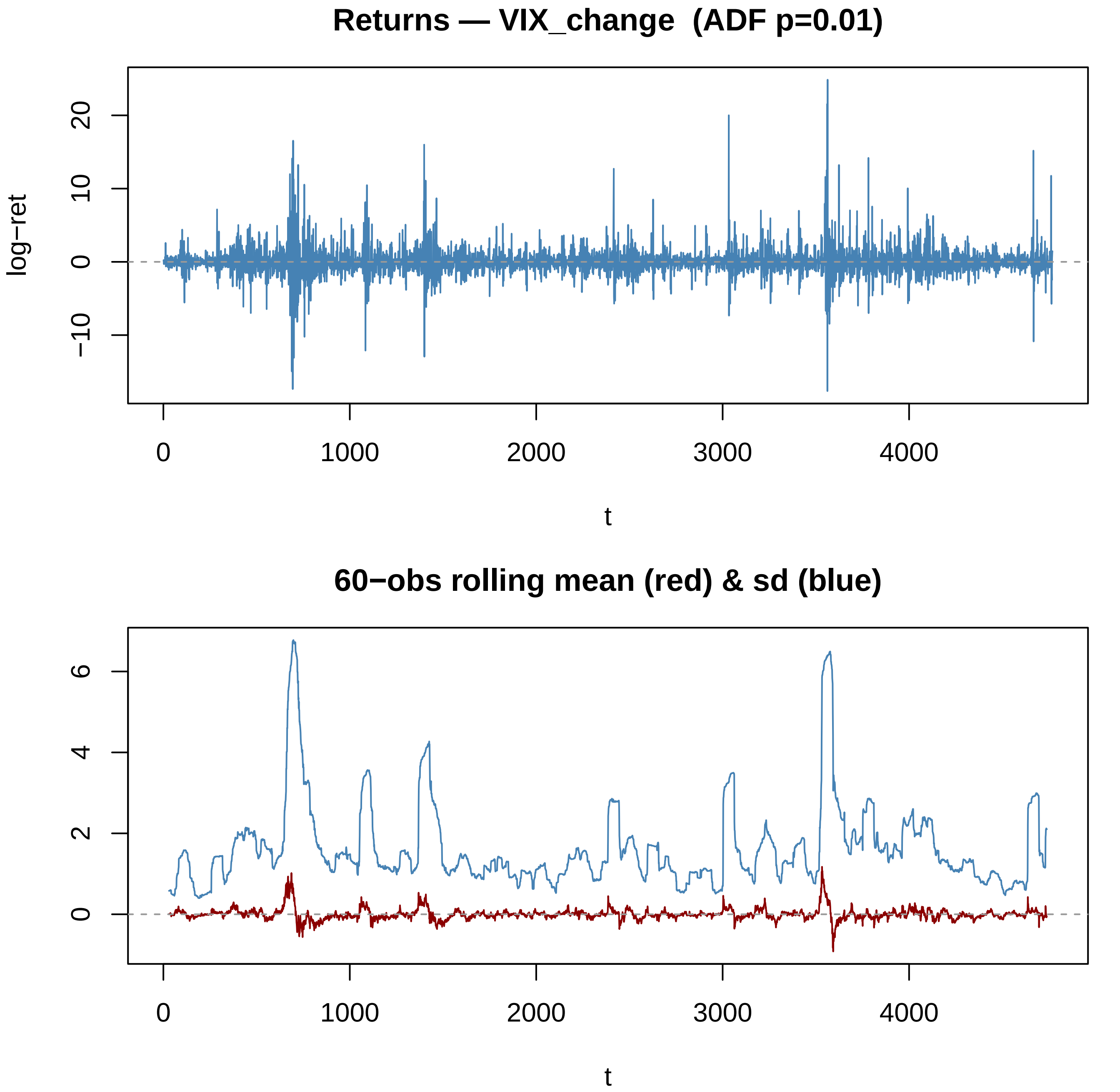


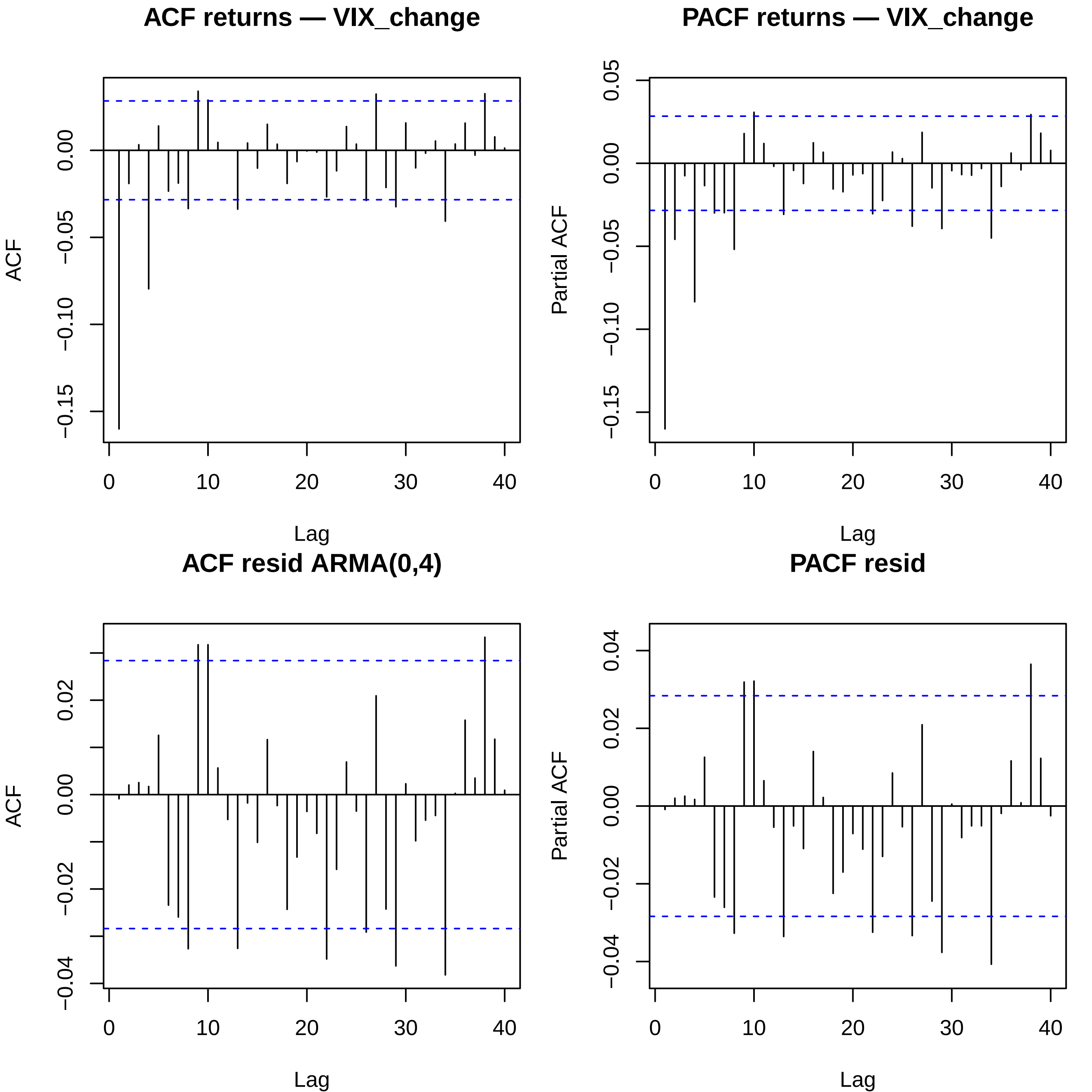
gjrGARCH(1,1)−nig | SPY_level_ch
Std residuals ._t
._t vs N(0,1)
ret
Density
ACF . (WLB p=0.05)
ACF .² (WLB p=0.907)
Variance ratio=0.99 dev=−0.5%
cond s^2
emp var
model uncond
ACF
s^2
Lag
t

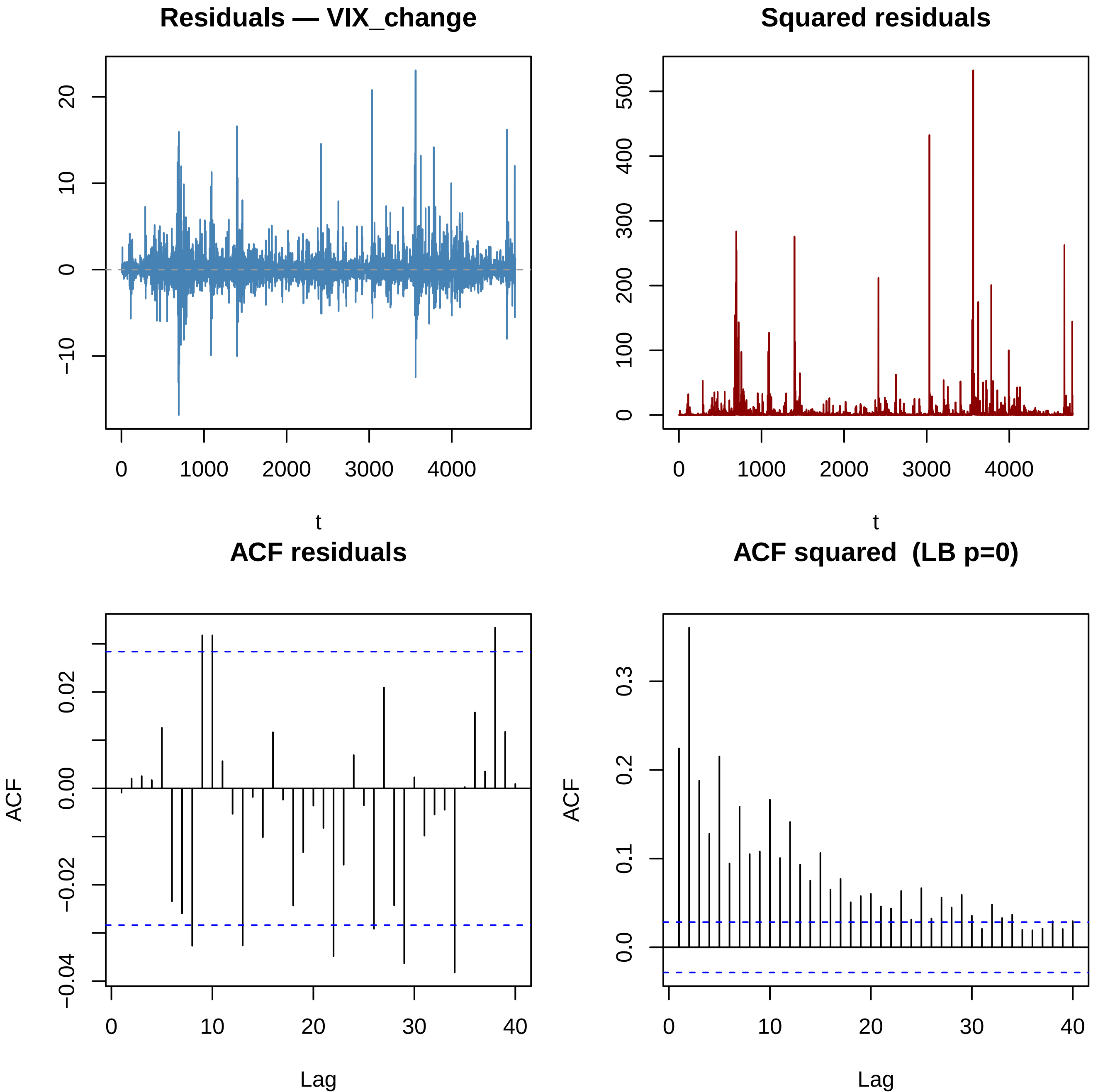
8 — PIT density fits to ._t — SPY_le




NO KS p=0
TF KS p=0.268
LO KS p=0.176
SST KS p=0.178
ST3 KS p=0.178
* JSU KS p=0.243
empirical
fitted

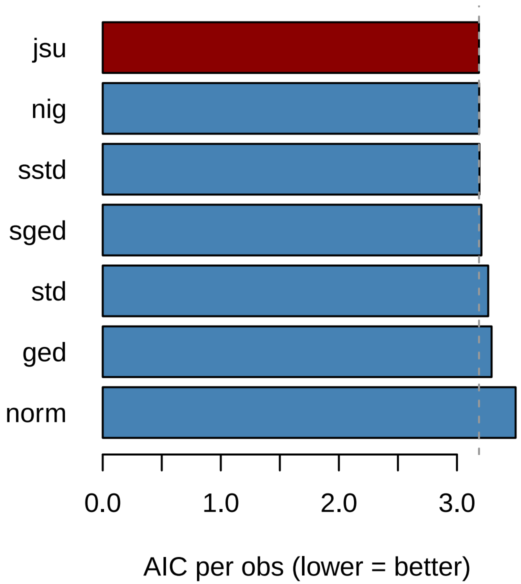

Returns — VIX_change (ADF p=0.01)
log−ret
20
10
0
−10
0
1000
2000
3000
4000
t
60−obs rolling mean (red) & sd (blue)
6
4
2
0
0
1000
2000
3000
4000
t

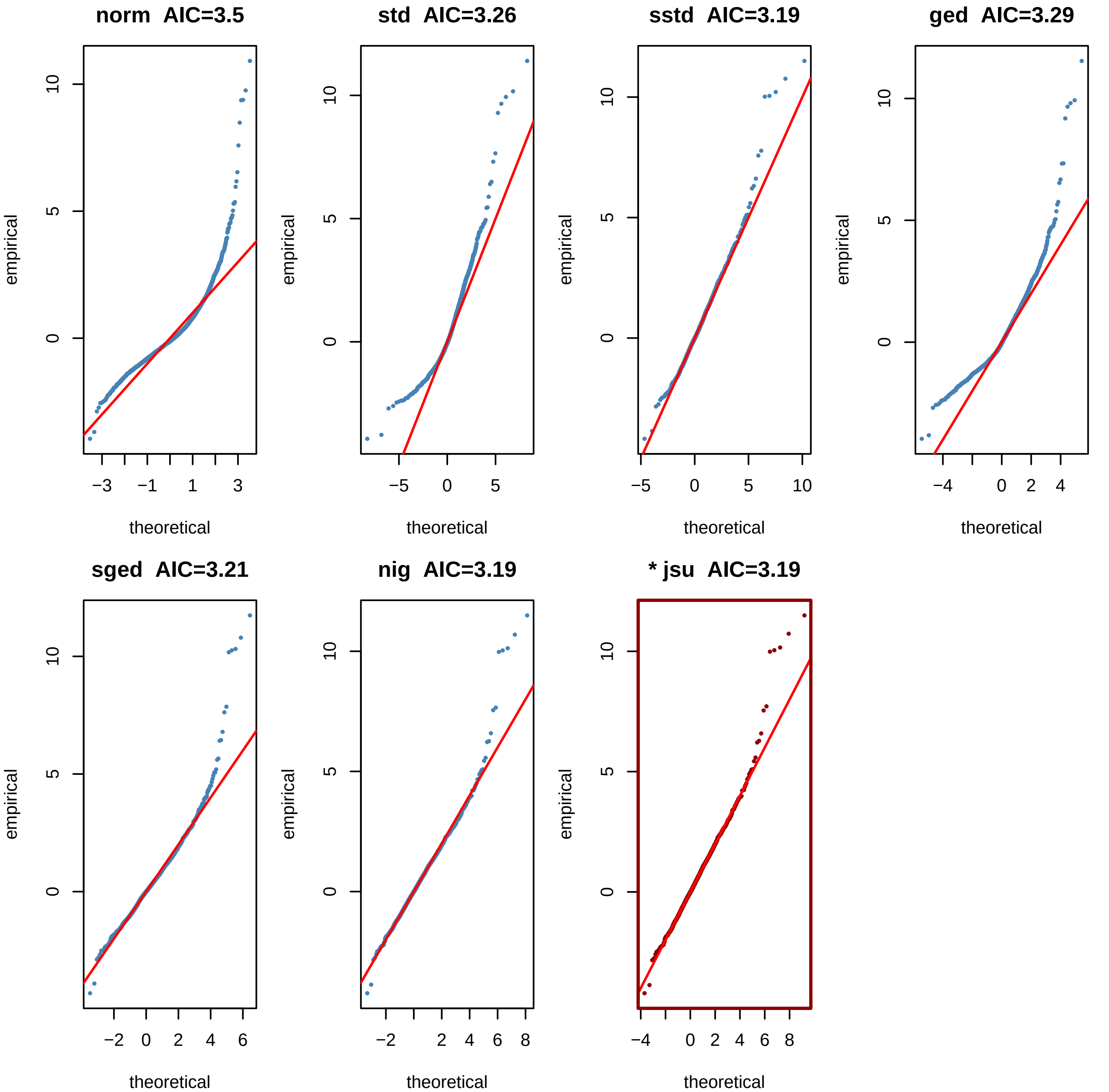

ACF returns — VIX_change
ACF
0.00
−0.05
−0.10
−0.15
0
10
20
30
40
Lag
PACF returns — VIX_change
Partial ACF
0.05
0.00
−0.05
−0.10
−0.15
0
10
20
30
40
Lag
ACF resid ARMA(0,4)
ACF
0.02
0.00
−0.02
−0.04
0
10
20
30
40
Lag
PACF resid
Partial ACF
0.04
0.02
0.00
−0.02
−0.04
0
10
20
30
40
Lag

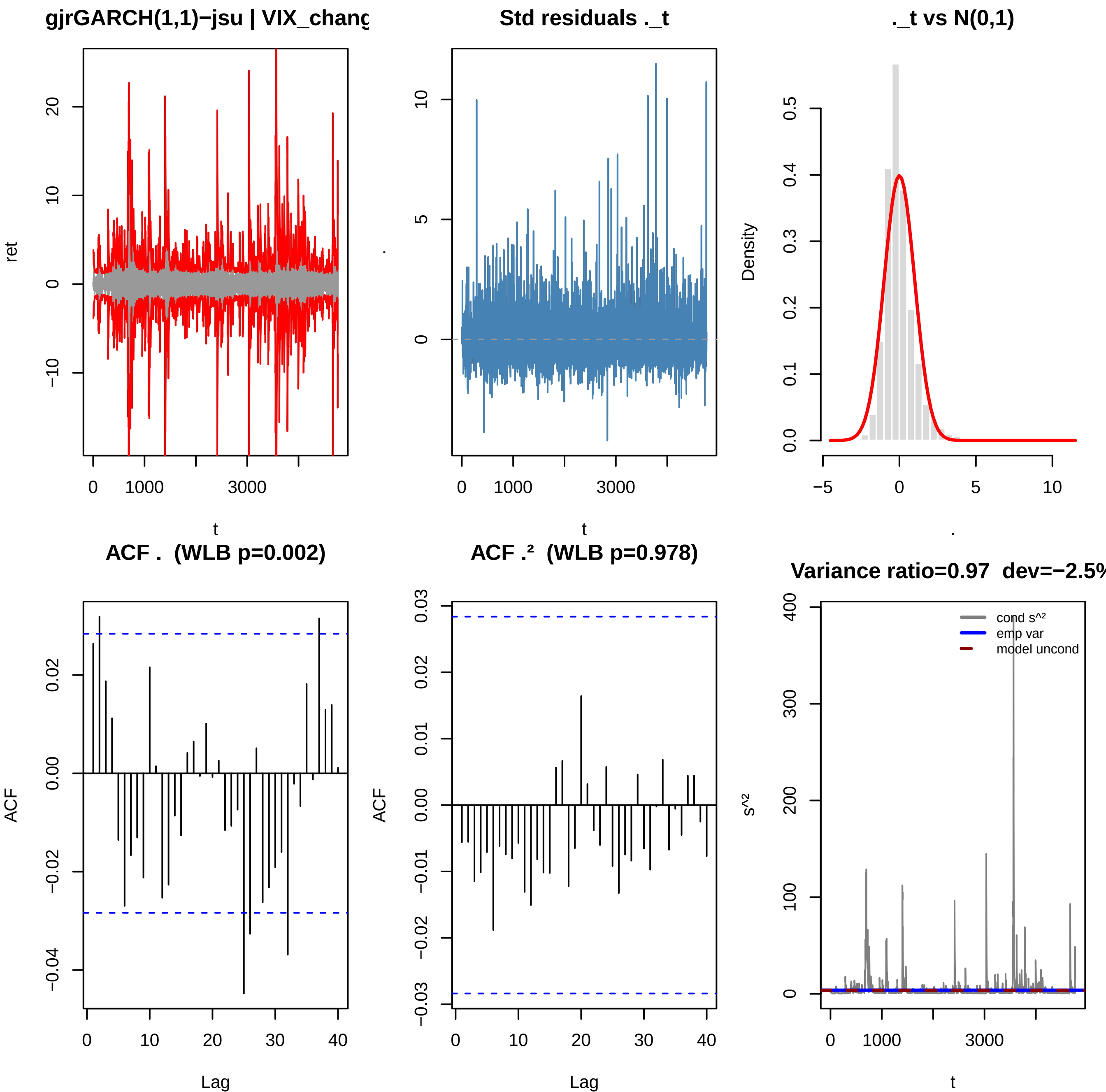


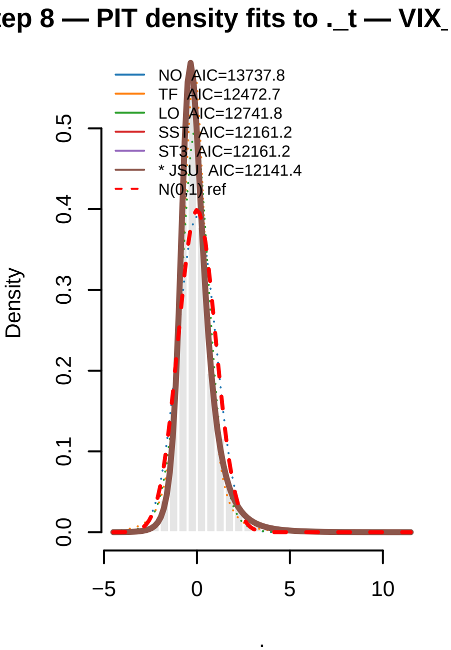
Residuals — VIX_change
20
10
0
−10
0
1000
2000
3000
4000
t
Squared residuals
500
400
300
200
100
0
0
1000
2000
3000
4000
t
ACF residuals
ACF
0.02
0.00
−0.02
−0.04
0
10
20
30
40
Lag
ACF squared (LB p=0)
ACF
0.3
0.2
0.1
0.0
0
10
20
30
40
Lag

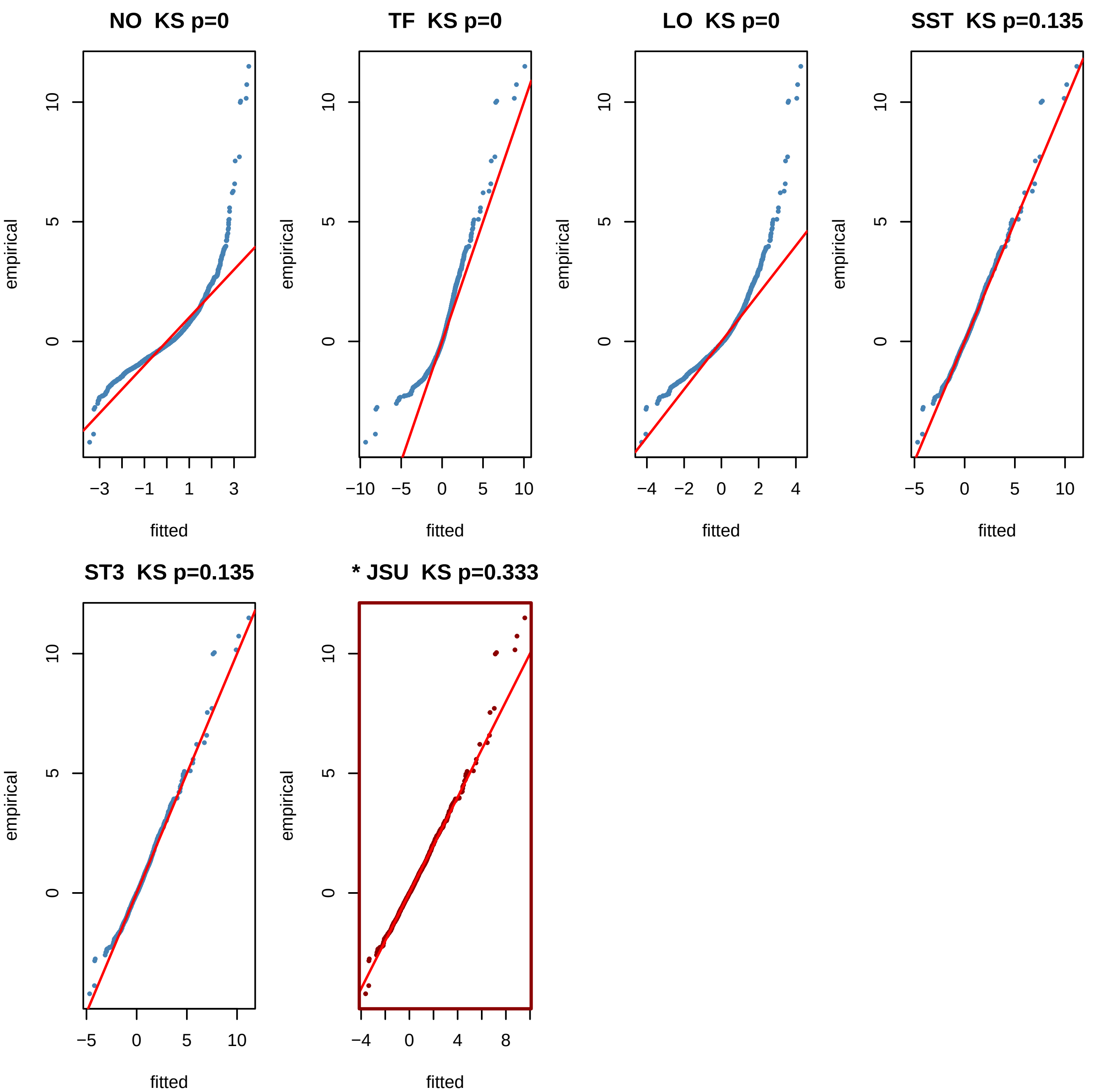
Step 6 — GARCH innovation dist AIC — VI




norm AIC=3.5
std AIC=3.26
sstd AIC=3.19
ged AIC=3.29
sged AIC=3.21
nig AIC=3.19
* jsu AIC=3.19
empirical
theoretical


gjrGARCH(1,1)−jsu | VIX_chang
ret
t
Std residuals ._t
t
._t vs N(0,1)
Density
ACF . (WLB p=0.002)
ACF
Lag
ACF .² (WLB p=0.978)
ACF
Lag
Variance ratio=0.97 dev=−2.5%
cond s^2
emp var
model uncond
s^2
t


ep 8 — PIT density fits to ._t — VIX_
NO AIC=13737.8
TF AIC=12472.7
LO AIC=12741.8
SST AIC=12161.2
ST3 AIC=12161.2
* JSU AIC=12141.4
N(0,1) ref
Density
0.0
0.1
0.2
0.3
0.4
0.5
−5
0
5
10


NO KS p=0
TF KS p=0
LO KS p=0
SST KS p=0.135
ST3 KS p=0.135
* JSU KS p=0.333
empirical
fitted